

The Czech retail industry work environment changes in the context of the fourth industrial revolution.

Diploma Thesis

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Statutory declaration

I, Scott Rockson, hereby declare that, this thesis entitled: **The Czech retail industry work environment changes in the context of the fourth industrial revolution**, was written and completed by me. I also declare that all the sources and information used to complete the thesis are included in the list of references. I agree that the thesis could be made public in accordance with Article 47b of Act No. 111/1998 Coll., Higher Education Institutions and on Amendments and Supplements to Some Other Acts (the Higher Education Act), and in accordance with the current Directive on publishing of the final thesis.

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In Brno on December, 2023

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1 Introduction

1.1 Background of the study

In any industry, sector or organization, there must always be room for flexibility and adaptation to make room for unforeseen future changes, to and from every factor of production. Whenever there are changes in an industry such as machinery, human capital and innovations, these tend to be revolutionary and demands response from especially managers and the labor force. As it has been witnessed through the history of industrial revolutions, it has then become imperative to study, and focus on the changing nature of work and employment relationships. This is to say that economic developments are increasingly impacting on work and employment relations in terms of how work duties are being performed, how decisions are taken, the nature of working hours, work sharing, full-time or part time and many other work relations that have evolved over the years (Lansbury, 2018).

There is no denying that recent developments in technology and artificial intelligence is transforming jobs and the way of work as well as their processes. New technologies are reshaping the ways that goods and services are developed, produced and distributed. Machines are increasingly performing the work duties that humans used to do, and even though there are new jobs for people, these jobs mostly require technology oriented literacy and experience to perform those (Farrell et al., 2020). Altogether, it is important for managers and employers to take the lessons that came with the first industrial revolution and they must understand that the extent to which society accepts and works with technological innovation is a major determinant of progress (Schwab, 2017).

1.1.1 Fourth Industrial Revolution

The fourth industrial revolution, often called Industry 4.0 or 4IR, builds on the foundations of the previous three revolutions and is set to change society, lifestyle, and work like never before. The fourth industrial revolution can be described as blurring of boundaries between the physical digital, and biological world. This implies the

fusion of advances artificial intelligence (AI), robotics, the internet of things, genetic engineering, among other technologies (McGinnis, 2020). Prime examples of these include GPS route system, voice activated virtual assistant systems, smart homes, advertisement recommendations, to mention a few.

Just as the first, second and third industrial revolutions that came with marked transitions in our societies, way of life, and work, the fourth industrial revolution is also capable of improving the life standards and quality, as well as income levels across the world (Schwab, 2017).

However, the revolution has sides to it. The optimistic view of it such as the developments mentioned above, and then the pessimistic view of the 4IR, which could pose threats and job losses. According to Schwab, “The changes are so profound that, from the perspective of human history, there has never been a time of greater promise or potential peril” (2017, pg. 8). There is a view that the fourth industrial revolution could lead to inequality, as it has the potential to disrupt labor markets. The job market may be pushed into “low-skill/low-pay” and “high-skill/high-pay” roles, which would eventually exacerbate social tension, that is, people with higher skills will continue to earn more, and those with lower skills with respect to the demands of digitization will earn lesser incomes (Schwab, 2017).

1.1.2 Retail in Czech Republic

The Czech Republic (CR) retail market is considered to have one of the top proportions of supermarkets and superstore space per capita within all of Europe. According to Export Enterprise SA in 2021, commercial distribution in the Czech Republic was state operated until 1989. These dynamics have changed from only 1% hypermarkets and 49% traditional businesses in 1997, to the retail industry being completely dominated by supermarkets and/or hypermarkets. The current distribution structure of retail in the Czech Republic is characterized by big groups domination small local traders, just like the rest of Europe (Export Enterprises SA, 2023).

Czech has one of the highest proportion of promotional sales, while consumer expenditures on food and non-alcoholic beverages account for 20.7% of total household spending (Export Enterprises SA, 2023). Currently, German, UK and Dutch retailers dominate the retail market in Czech Republic. As of 2019, the top three retail groups were Schwarz group (Lidl), REWE group (Billa), and then Ahold (Albert), all holding up to 57% market share (Export Enterprises SA, 2023). It has been reported that the Czech retail market remains very attractive to external investors. In 2020, twenty-five new brands were introduced to the Czech market, and yet, the country has still recorded the arrival of thirty-eight new brands in 2021, being the highest in the last three years. This means a 52% year-on-year increase in the influx of retail brands (Property Forum, 2022).

2 Objectives

2.1 Aim of study

With respect to the background of the study, this research is conducted in order to assess the changes in the Czech retail labour market, which may come along with the technological advancements of the fourth industrial revolution and answer the set research questions which are listed in the chapter that follows.

To answer these research questions Microsoft Excel, Gretl and SPSS statistical softwares will be employed, where necessary, to conduct analysis on the data obtained from retail markets such as Billa, Tesco, among others.

This diploma thesis will be written in accordance with the set requirements for such an academic paper, with respect to the rules and guidelines set to make this research successful. The significance, and structure of this study are all outlined in the sub-chapters that follow.

2.2 Research questions

This study seeks to address these questions:

1. How will these technological advancements affect the jobs and the retail labor market?
2. To what extent will the evolution impact the future of this industry?
3. Why should the retail labor market embrace the evolution?

2.3 Significance of the study

The retail sector of the Czech Republic has seen major developments and continuous technological advancements that comes with the fourth industrial revolution, as well as innovations that sprung out of the inconveniences that came with the Covid-19 outbreak. There has been a soar in online purchases and delivery services, self-checkout panels, as well as the idea of fully automatic stores, operating around the clock without any human assistance or attendants after-hours (Adamcova 2022). All

these developments have come as a result of embracing technology and the fourth industrial revolution into the retail sector, and at the same time addressing issues in response to the changing tastes, preferences and methods of the everyday consumers.

Export Enterprises SA, (2023), in their publication referred to statements from Jan Kotrbacek, Partner and Head of Central and Eastern European (CEE) Retail Agency team, which elaborated on the fact that even though the Coronavirus changed the way of trading and posed challenges, physical shopping will always be important part of Czech retail. He further mentioned that retail in Czech is very important in Central and Eastern Europe, as it is open to foreign brands who count on the patronage of international visitors (Export Enterprises SA, 2023). There have been studies talking about the fourth industrial revolution and the future of work in general, but not much have however been researched on the effects, which the fourth industrial revolution brings to the retail labor market and it's environment. This research therefore attempts to elaborate how the changes associated with the fourth industrial revolution affects workers in the retail sector in Czech Republic. The research aims to propose recommendations that could be used to manage the impacts of these changes on the labor market. This research also has the ability to be used as a tool for future research and decision making in these kinds of strategic changes and the effects they may pose on the workers.

2.4 Structure of study

This work shall be comprised of seven major chapters. Chapter one is made up of the Introduction, mainly the background of the study. Chapter two talks about the main objectives of this work, the aim, research questions, significance and the structure of the study. Chapter three comprises the literature review of various works and books in relation to this topic. Chapter four will spell out the research methodology the researcher adopted to undertake the study as well as the method of data analysis. Chapter five entails analysis and interpretation of the findings that resulted

from carrying out this research. Chapter six deals with proposed solutions and recommendations that could be used to manage findings of the research. Chapter seven will discuss possible shortcomings of the work, followed by conclusions of the work.

3 Literature Review

This chapter aims to provide a review of existing literature which have relations to this study. The literature review looks at books, journals and theoretical literature that explains relevant subjects for this study as well as considering empirical research that has been conducted concerning this study topic.

3.1 The history of industrial revolutions

Revolution in itself, can be said to be a sudden or drastic change in an entire social order or organization, affecting all phase of the order. It can be said to be the obstruction and/or at the same time, the emergence of social structure (Yoder, 1926). Revolution could be associated with many facades of society, be it political, industrial, economic, religious, among other social aspects. Revolutions have been known to occur through the history of mankind and that has resulted in shifting our way of living, and has been perceived to be usually triggered by new technologies and new perspectives on living.

In view of this, Industrial revolution can be defined as the absolute growth of industry and its expansion relative to other sectors of the economy and all its respective sub-sectors, which has been seen as the basis of modernity, various innovations and transformations in the world (More, 2002). Prior to the industrial revolution, most of the world's economy had predominantly been about agriculture and animal farming, trading and crafting, such as blacksmithing, weaving, and so forth. Since then, the world as we know has seen four industrial revolutions, the fourth in which we currently live (Yildirim & Çeştepe, 2018).

The first industrial revolution which started in England, spread across all of Europe and then eventually to America, occurred within the 1760s through to around 1840, based on water, steam and mechanical production machinery. This is said to be the premiere shift in our way of living, transitioning from muscle power, to mechanical power. This era was propelled by the invention of the steam engine, spinning machines, looms for fabric production, as well as using coal and steam as fuel

instead of wood, which resulted in massive extraction of coal to run and provide this new energy form. This revolution ignited production through machine use, as opposed to muscle strength before it, thereby changing the means through which goods were being produced and exchanged. For the first time in history, industry started to overtake agriculture as the backbone of economies of the world. Economies became accelerated, railroads were invented, hence amounting to increase in production, productivity, and transportation. Not only that, the standard of living was improved, average lifespan prolonged and production in big factories started to rise (Yildirim & Çeştepe, 2018).

In the late 19th century, the second industrial revolution was birthed, often themed as the most important industrial revolution which spanned right into the early 20th century. This era was characterized by the discovery of electricity, oil and gas. These energy sources mentioned eventually resulted in the creation of combustion engine, which has been used until recent times (Schwab, 2017). The electric light bulb was invented for individual households within this revolutionary period, in 1879, which changed the way of life and work of the people, before which most tasks had to be performed when there was daylight. These inventions are the reason this era is named as the most important, not forgetting the telegraph and telephone modes of communication, as well as invention of automobiles and cars, which have been improved but still used in today's world. In the period of the second industrial revolution, mechanical production sped up, resulted in mass production and exploration of new types of fuel and energy (Willems et al., 2018).

The third industrial revolution started between the 1960s and 70s. This era was dominated by the use of newer technologies such as computers; mainframe in the 1960s and personal computers in the 1970s, the internet world wide web, microelectronics, telematics, among others. This revolution which still stands as the foundation for most development we have today, is also termed as the digital or computer revolution (Schwab, 2017). In this era of the revolution, inventions like semi-

conductors, mainframe computers, and analogue television were introduced together with the use of complex manufacturing processes which disrupted industries and boosted the growth and globalization of supply chains (Janicke & Jacob, 2009).

Industrial revolution therefore can be said to not be, even though is, such an abrupt incidence but series of developments that occur over time unlike the literal meaning of the word revolution. In literal view, a school of thought might argue that an industrial revolution must be evident through discontinuity in economic activities as well as indicators. It has however been seen that economies saw widespread innovation across various sectors, which revolutionized the world in our ways of living, production, trading, travelling, working, and every social façade of life. Industrial revolutions can be said to be a build up from the previous developments, leading our world into a space of perception in terms of structures of the society and economic systems (Hudson, 2014)

It is then imperative to understand the waves of industrial revolutions that have come before today and paved the way, for the revolution we live in at this time. Knowing how it worked, the way it changed society and the way of life and work, will help us understand the age and technology we have today, how it has happened and the possible impacts it may bring. As digital technologies are still in use and being developed, almost as though we are just living in an upgrade of the previous industrial revolutions, we need to understand the fourth industrial revolution of today, what it entails, and where it may lead.

3.2 The fourth industrial revolution

We are currently living in the era of the fourth industrial revolution, also known as Industry 4.0, which describes the current trend of automation and data exchange in technologies. This revolution have had so many scholars define it in many different ways and with many different perspectives, but every view on the industry 4.0 nonetheless, points to the same analogy in the end, Digitization and Integration.

The Fourth Industrial Revolution, was termed as Industry 4.0, for the first time during the “Hannover Fair” in 2011, led by the Ministry of Education and Research

(BMBF) and the Ministry for Economic Affairs and Energy (BMWi), with the main objective to advance digital manufacturing by promoting the digitization and interconnectedness of products, value chains, and business models. The communication Promoters Group of the Industry Science Research Alliance (FU) launched the initiative in 2011, aimed at facilitating research to improve the means through which they manufacture things, engineering systems as well as how to manage the changes in all sectors. The Industry 4.0 initiative was then adopted by the High-Tech Strategy 2020 Action Plan (Conrads et al. 2017). Since then, the initiative has helped to bridge the gap between different industry sectors, facilitated the smooth transition and implementation of research into actual industrial, and has seen reasonable growth to become a very important milestone towards consolidating Germany's technological leadership in mechanical engineering. This initiative was then popularized by Klaus Schwab, founder and Executive president of World Economic Forum in 2016 as the 'Fourth Industrial Revolution', which has now been adopted and used all over Europe, the Americas, Great Britain, and all over the world (Postelnicu & Câlea, 2019).

The fourth industrial revolution could be defined as a merge of physical assets and advanced digital technologies that analyze, communicate and act upon information available, to enable individual consumers, organizations and society as a whole to be more flexible, responsive, and make more data-driven decisions intelligently and with precision (Deloitte, 2020). These digital technologies include drones, Artificial Intelligence (AI), robots, self-driving vehicles, 3D printers, nanotechnology, cloud computing and a whole lot more. The fourth industrial revolution has often been talked about as the Industrial internet of things, which has been described that, it "provides a way to get better visibility and insight into the company's operations and assets through integration of machine sensors, middleware, software and backend cloud computer and storage systems" (Gilchrist, 2016).

According to Klaus Schwab (2017), what sets the fourth industrial revolution apart from the previous ones is the way these technologies have been merged and how they work together with each other. This integration has resulted in faster and wider spread of emerging technologies and innovations. The interaction across

physical, digital, and biological domains has created a new level of connectivity that has never been seen before. While previous industrial revolutions were characterized by the introduction of new machinery and automation, the fourth industrial revolution encompasses much more. In addition to use of smart devices and connected machines, it also includes ground-breaking discoveries and inventions in various fields such as gene sequencing, renewable energy, and quantum computing. However, it's worth noting that while some parts of the world are fully embracing the fourth industrial revolution, others are still living in the second or third industrial revolution. For instance, in some parts of the world, farming is still carried out using brawn strength without machinery (Nchuchuwe & Adejuwon, 2012). Additionally, there are still people living with little or no access to the internet and computers. Nevertheless, the potential of the fourth industrial revolution to spread at an unprecedented pace is clear, as seen in the rapid adoption and utilization of the internet, social media marketing, and usage all over the world. The pace of technological advancement and innovation is expected to continue accelerating in the coming years, with implications for both developed and developing countries.

3.3 Drivers of the fourth industrial revolution

As mentioned in the previous chapters, all previous industrial revolutions were based on discoveries that changed social systems and way of life. The fourth industrial revolution is no different from the older three revolutions, as its spread and potential stands on drivers, also known as Megatrends, according to Schwab K., 2017.

These are the features that have enhanced and enabled the fourth industrial revolution through the adaptation, digitization and utilization of modern technological developments. In other words, what are the main things that stand as evidence of the fourth industrial revolution? We shall discuss the Megatrends, according to the fourth industrial revolution written by Klaus Schwab in three main clusters: Physical, Digital and Biological, to help us understand the depth and fundamentals of fea-

tures of modern technology that shows the direction of this current industrial revolution (Schwab, 2017). Despite the fact that these cluster of features are each distinct and stand on their own, their interdependence and how intertwined their functions could be is key to examining the benefits, effects and the nature of the fourth industrial revolution.

3.3.1 Physical Drivers

The group of physical aspects of the megatrends are the tangible parts of the world's transformation technologies. These prevalent and fascinating physical manifestations include Autonomous Vehicles, Advanced Robotics, New Materials and Three-dimension (3D) Printing.

- **Autonomous vehicles** (driverless cars, drones and boats) are rapidly advancing in their capabilities, thanks to advancements in technologies relating to sensors and artificial intelligence. As some cars can travel driverless and avoid obstacles and crushing into traffic, drones, in the same way could be used in situations like, delivering goods to remote, inaccessible and deprived areas, and they will be able to execute these functions perfectly using the data available and artificial intelligence, learnt from the self-driving cars. These functions could also be enhanced to benefit agriculture by using data analytics in combination with drones to enable efficient and precise farming practices like the exact time plants need to be watered, the exact amount of agro-chemicals they can tolerate, and the application of all other supplies to meet the needs of agricultural products to improve yield (Schwab, 2017).
- **Advanced robotics**, thanks to progress in sensors and information technology are increasingly learning and becoming flexible in their use and functions. This encompasses the possibility to enhance and emphasize how humans and machines can work together. For example, underwater robotics can be used to reach depths that could be dangerous to human divers, to observe and facilitate scientific studies of the marine world (Yuh & West, 2001). With the help of cloud technology, and the ability to access information remotely, advanced robotics

can collaborate to work with precision in fields like surgery, nursing, agriculture, simulation training and drills, amongst others, just by learning from how human specialists perform these activities as they work together.

- **New materials** that have not been in use ever before, could be enhanced and utilized in, and by the advancements of the fourth industrial revolution. These new materials are considered to be way better substitutes to those that are currently in use, and with future improvements, will continue to be much lighter, stronger, recyclable, and adaptive, which could be applied as smart materials. The nanotube, which is a narrow and very tiny tube with a diameter of one nanometer (1 nm) developed from fullerene carbon structure graphene is one example of these new materials. Nanotube, which could serve as strength reinforcement in structures as well as signal transmitters is considered to be one-sixth the weight of steel but however possess at least 100 times the strength of steel, and is also considered a very good heat and electricity conductor, far better than copper (Lau & Hui, 2002). In the case where nanotubes are then produced in large quantities and become price competitive, it could significantly disrupt manufacturing and infrastructure industries that produce and depend on steel, as well as countries without the scientific capacity to match these improved new materials for production.
- **3D printing**, also known as additive manufacturing is revolutionizing the way objects are created. 3D printing is a form of manufacturing where materials like plastic or metal are stratified to build objects into a three-dimensional shape such as sunglasses, artificial teeth, engineering prototypes, and others (Schubert et al., 2014). Any word lesser than ground-breaking cannot describe the capabilities of this revolutionary invention. 3D printing still has the potential to be used in schools for educational purposes, factories for making product models, in architecture and many other fields for more widespread use once they have been optimized to smaller and manageable size and cost efficient. Currently, research is being done to print smart materials that allows for self-repair, self-assembly and require stimulus-responsive materials to make them more advanced than,

but based on the technology of the 3D printing we have today, termed as 4D printing (Momeni et al., 2017).

As the physical world continues to transform at a rapid pace, it is clear that these technological megatrends will have a profound impact on society, the economy, and the environment.

3.3.2 Digital Drivers

The digital group of megatrends explores the virtual bridge that joins the physical and virtual world together in the fourth industrial revolution, with the main focus on Internet of Things (IoT), Blockchain Technology, and on-demand economy.

- **The internet of things** “provides a way to get better visibility and insight into the company’s operations and assets through integration of machine sensors, middleware, software and backend cloud computer and storage systems” (Gilchrist, 2016). The fast spread of the use of sensors and other means of connecting physical objects, places and objects to virtual networks has made it possible to monitor and optimize assets, people, animals, as well as industrial spaces. An example of the transformative applications of the internet of things is remote monitoring, where companies can track packages as they move through the supply chain in real time. It can also be used to track aircrafts and ships, in case they miss their course or go missing due to unforeseen circumstances, improving accountability and satisfaction of the final consumer.
- **The Blockchain technology**, originally conceived as the foundations to cryptocurrencies describes the technology that handles blocks of uniquely identified transaction records in chain. It uses a decentralized record of transactions between contracting parties which independently verifies transactions before applying them, secured by cryptographic sealing (Treleaven et al., 2017). The transactions are then bound by a smart contract, rules that the involved parties have agreed upon to govern their transactions. The extension of the use and application of the Blockchain technology could enhance secure storage and exchange of information, digital registrar for things such as biodata, using the

smart contract to automate terms and laws to improve transparency and accountability.

- **The on-demand economy** refers is said to be made up of digital platforms that connect service agents with waiting customers in a situation that is normally time-sensitive, thereby creating new ways of consuming goods and services (Taylor, 2018). These platforms are basically intermediaries that intend to fill the needs of the users involved where they want it and when they want it. These digital platforms are not limited to use, but are spread across various sectors of the economy for example Retail (Rohlik & Wolt), Transportation (Bolt), Fashion (Shein), Delivery (PPL), and many more. The on-demand market responds to the demands of the consumers from the comfort of their homes or wherever they may find themselves. Now, a user can hail a ride from their homes in just a few seconds instead of trekking to a bus/train station, users can just order food on these platforms instead of going to the restaurant or grocery shops, and not forgetting the traditional way where the malls connected consumers with different markets, now people can just order their clothes, electrical gadgets, and even the services of specialists like plumbers and carpenters from an application (app) on their phones and computers (Delgosha & Hajiheydari, 2020). The comfort and the abundance of these apps have actually created a digital economy of its own, with minimal charges than it may cost its users if they went for the goods or services on their own, saving them money, time and unnecessary distractions, hence enabling users to channel their attention to some more pressing needs. “Digital platforms have dramatically reduced the transaction and friction cost incurred when individuals or organizations share use of an asset or provide a service” (Schwab, 2017).

3.3.3 Biological Drivers

The technological integration into our way of life in the fourth industrial revolution does not have a further effect on us more than its impact on our biology (Genetics and Deoxyribonucleic Acid (DNA)), which are the basic building blocks for humans

and all other organisms (Britannica, 2023). In the year 1990, the Human Genome Project (HGP) was rolled out to develop the genetic sequencing of the human genome by 2005, with investments from countries like the United States of America (USA) the initiator, Britain, Germany, China, Japan, France and Canada, which took 13 years to complete. This led to the discovery that 99.9% of human DNA sequences are the same and the other 0.1% of variation was responsible for the genetic risk of diseases and illnesses (Collins & McKusick, 2001). Fast forward to 2012, this discovery led to the production of an approved drug to treat a group of patients with cystic fibrosis, using the genome variation discovery to develop tailored drugs. According to Gibbs, R., 2020, the Founding Director of the Baylor College of Medicine - Human Genome Sequencing Center (HSGC), human Genome sequencing now costs less than United States Dollar (USD) \$1000 per genome, while a broad stream of biological data are made readily available and easy to analyze for scientists through the availability of powerful computers, the growth in computational biology, and genetic genome map information. With the growth in technology and scientific advancements, these researches could be used find out the foundations of inherited disorders and get answers to diseases like cancer. Genetic information has also been made accessible for people who want to know about theirs, to provide predictive possible illnesses, as well as providing personalized therapy to make sure that individuals stay as healthy as possible, and do things that will give them a stronger chance to fight any unwanted genetic issue (Collins & McKusick, 2001). Not only limited to genetic sequencing and editing, another biological advancement is the Assisted Reproduction Technology (ART) used for screening unborn babies to improve healthy conceptions, provide solutions to miscarriages, and also detect possibility of abnormalities in embryos before initiating conception (Cervero et al., 2013)

The potential applications of biological technology has endless possibilities and applications in research into other animals, bacteria, viruses, as well as food and vegetation. In any case, there will always be the chance for scientists to explore the possible knowledge pool to know more about human nature, and how to deal with difficulties, ailments and deformities that may arise, and find befitting solutions to

fight them, whether specifically tailored to suit a certain type of genetics, or a widely accepted clinical solution to health problems across the world.

3.4 Impacts of the fourth industrial revolution on labour

The rise of artificial intelligence, the internet of things, robotics, big data and other key drivers of the fourth industrial revolution are already transforming the economic, physical and cultural aspects of the world in various ways (Schwab, 2017). These transformations have been a topic of much debate when it comes to the labor market, with some arguing that it will create new jobs, while others believe it will lead to job displacement which have been of growing concern in the world today (Goos et al., 2019). This chapter seeks to review some ways in which the fourth industrial revolution and the integration it comes with, does or could impact the way we work and how this transformation could affect the labor market.

- **Work structure and organization.**

According to Goos et al. (2019), the Fourth Industrial Revolution (4IR) has brought about a need for various institutions to adopt new technologies in order to remain competitive and maintain quality production or services. However, implementing these technologies may require changes in organizational structure and processes, which can have significant implications for the labor market. The adoption of new technologies of the fourth industrial revolution, such as automation and robotics, are being used to replace repetitive or dangerous tasks previously performed by humans and this can result in job losses and potential displacement of workers (Yuh & West, 2001). Furthermore, adjusting the workplace to meet the growing technological world of work may require the scrapping of current structures and processes, potentially affecting the labor force in these fields. This may include taking away structural hierarchies and levels of management, creating new work groups or dismantling the current structure altogether. Changes in organizational structure can lead to uncertainty among workers about their future, which could negatively impact their confidence and productivity, even

though they were implemented with the intention to increase productivity. Schwab, K. (2017), argues that this may cause anxiety and stress among workers who are uncertain about their role within the organization. This can lead to a decrease in productivity and efficiency, which may ultimately affect the success of the organization.

- **Skillset.**

The fourth industrial revolution has introduced new forms of work in fields such as data analysis, digital marketing, artificial intelligence, cybersecurity, and many more. This means that, with new forms of work, comes with the need to adjust and attain skills that will enable the labor force to work efficiently (McGinnis, 2018). To adapt to the changes in the labor market, employees will more likely be required to improve their skills in their current jobs or learn skills for totally new jobs to equip them with competent skills for the future. Even though there exists the likelihood of people having to gain more skills to be able to stay relevant in the field of work, automation and use of robots in some fields might actually see a decrease in skill requirements (Goos et al., 2019). Then again, the changes that come with the fourth industrial revolution may have lesser impacts on non-routine jobs. This, according to Schwab, K. (2017), is may be due to the fact that automation may be implemented in the tasks that are repetitive and systematic, hence leaving room for jobs that require soft skills and creativity, hence, heightening the focus of soft skills for some jobs, and technological know-how, for others.

- **Working relationships**

With the advent of digital platforms and technological advancements, there will be a high likelihood that even more and more workers will become freelancers, working on project-to-project basis, instead of holding a permanent employment. This implies that the nature of employment relationships between workers and employers will massively be impacted (McKinsey Global Institute, 2017). As a result of technological advancements and changing traditional working relationships, workers will tend to shift towards more flexible, fluid and temporary work with increase in remote work, gig economy and collaboration.

According to “The future of jobs report” by World Economic Forum, 2018, recent advancements in technology have brought about significant changes in the way businesses operate, by seeing automation become an increasingly popular trend, as companies seek to streamline their operations and increase productivity. A survey from the aforementioned report mentions that nearly half of all companies involved in their survey anticipate that automation will lead to a reduction in their full-time workforce by 2022, based on their current employee base, thus highlighting the increasing trend towards a more flexible and decentralized workforce. Many companies expressed their intention to utilize remote staffing beyond physical offices, and to engage workers in a more flexible manner. This includes an expansion in the use of contractors doing task-specialized work, indicating a shift towards a more agile and adaptable workforce, hence influencing companies to be more agile and responsive to the swiftly changing work environment and adopt appropriate staffing strategies.

- **Labor substitution.**

As machines become more intelligent and capable, they can perform specific tasks with greater accuracy, speed, and efficiency than human workers. This means that companies are likely to increasingly turn to automation to replace human workers in certain jobs, particularly those that are low-skilled and require repetitive manual labor. The World Economic Forum's "The future of jobs report", 2018, revealed that a significant proportion of businesses are optimistic about the potential benefits of automation as 38% of respondents reported that they plan to extend their workforce to new productivity-enhancing roles. Also, over a quarter of businesses expect automation to lead to the creation of new roles within their organization, which implies that even though implementation of automation may result in job displacement for some workers, it also presents new opportunities for businesses to innovate and improve their operations. What this will mean for the labor force is that employees will need to be self-reliant and be able to effectively market their skills and services, and be willing to learn new skills and work for different clients and new roles as well (McKinsey Global Institute, 2017).

According to a report by McKinsey Global Institute, (2017), it is estimated that up to 375 million workers (or roughly 14% of the global workforce) may need to switch occupational categories and learn new skills by 2030 due to automation. However, it's important to note that not all jobs are equally susceptible to automation. For example, jobs that require complex and varied tasks, as well as those that require high levels of social and emotional intelligence, are less likely to be automated.

- **Quality of life.**

To talk about the impact of the fourth industrial revolution on the labor force, is to talk about how it would impact their work-life balance, which is a more direct effect on employees' way of life and work and the relationship in-between. One aspect of the fourth industrial revolution is that it may allow for autonomy and flexible style of work on the side of the worker, and this flexibility enables the labor force to better balance work and family responsibilities (Schröder, 2016). In this case, there is the potential of expectations being placed on the worker to be available and respond to work-related communications outside the traditional working hours which have been removed by the development of industry 4.0 and hence removing any boundaries that may exist between private life and work. Schröder, C., (2016), goes on to argue that digitization and flexibility in the working environment helps in data collection and data management and by analysing data on how workers move and perform tasks, employers can make adjustments to work processes and equipment to minimize physical strain and reduce the risk of injury. This can ultimately improve the health and well-being of workers, and contribute to a better work-life balance.

Kusmin, K. L., (2018), also argues that since the fourth industrial revolution comes with the emergence of new business models and significant changes in the workplace setting, automaton may be set to undertake repetitive tasks which will imply that the labor force can focus more on complex and challenging tasks in order to amplify employee abilities to manage their workload. This will have the potential to improve the quality of work and lives of workers, and increase their inspiration to innovation and productivity.

3.5 Industry 4.0 in Czech Republic (The Concept of Prumysl 4.0)

In September 2015, during the 57th International Engineering Fair in Brno, Czechia, the Ministry of Industry and Trade (MIT) presented a concept called Prumysl 4.0, which translates to Industry 4.0 in English. The initiative was introduced as a national plan to help the Czech Republic prepare for and increase its competitiveness in the face of technological advancements. It aimed to support the industries and the economy as a whole in adapting to the changes brought about by the fourth industrial revolution. The concept was officially approved by the government in August 2016. The goal of Prumysl 4.0 was to help Czechia realize its full potential while navigating the economic and social changes associated with the fourth industrial revolution. The initiative aimed to assist businesses in adopting digital technologies and to promote innovation in manufacturing processes. The implementation of Prumysl 4.0 was expected to bring about significant improvements in productivity, efficiency, and competitiveness across various sectors of the Czech economy (Adámek, 2018). The main objectives of the Prumysl 4.0 initiative were defined as below (Klitou et al., 2017).

1. To enhance the ability of Czech companies to participate in global supply chains.
2. The implementation of the Industry 4.0 principles will lead to efficient manufacturing, meaning faster, cheaper and resource-effective production.
3. To enhance the cooperation with R&D and industrial association, universities and the Academy of Sciences of the Czech Republic for the development of software solutions, patents, production lines and export know-how.

It is thus imperative for the country to ensure that its processes and sectors of the economy are ready to embrace and incorporate the methods and developments that come along with the fourth industrial revolution into their operations.

According to data on Statista, a third of the distribution of Czech Republic's Gross Domestic Product (GDP) across economic sectors is accounted for by the Industrial sector. As shown in Figure 1, an average of 32% of the GDP distribution by sectors, from 2011 to 2021 is contributed by the Industry sector per annum.

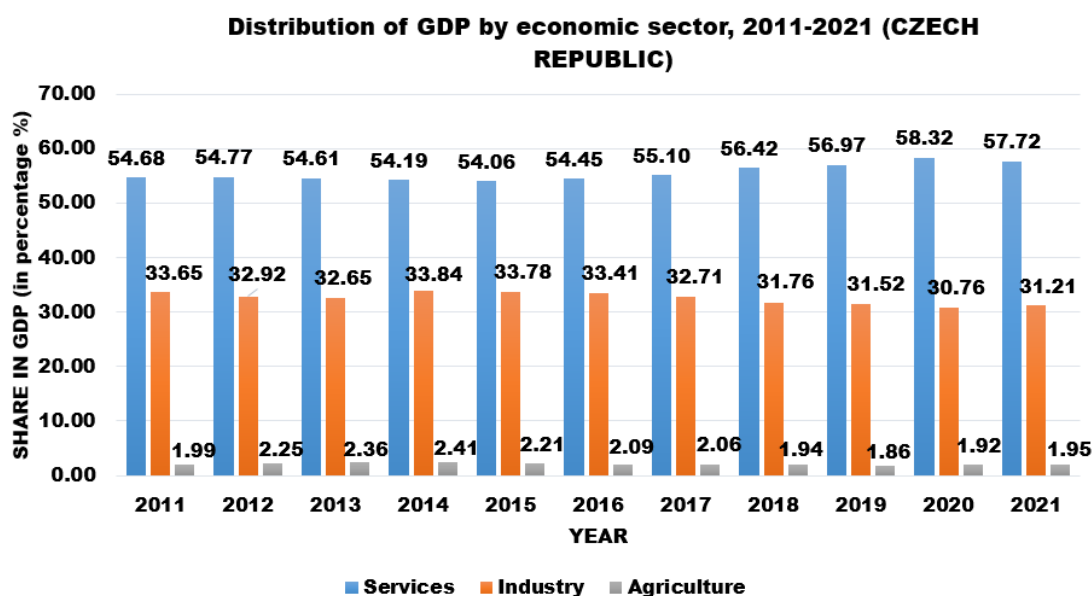


Fig. 1 Czech Republic: Distribution of Gross Domestic Product (GDP) across economic sectors 2011 to 2021. Source: O'Neill (2022). Statista.

Statista also published statistics that show the annual confidence index for the industrial sector in the European Union (EU) from 2008 to 2022. This index simply measures expectations based on surveys on a specific sector's performance, where negative values of the index indicates decline, and positive values indicate growth in confidence of that sector. This publication displays the aggregate confidence index for the member states of the EU industrial sector from 2008 – 2022, of which the statistics from 2018-2022 were projections by the database. As per these projections, the confidence index in the EU industrial sector indicated continuous confidence growth in their index points from 2017 - 2022, showing the potential growth of business confidence in EU industrial sector, and this is shown in Figure 2 (Statista, 2018).

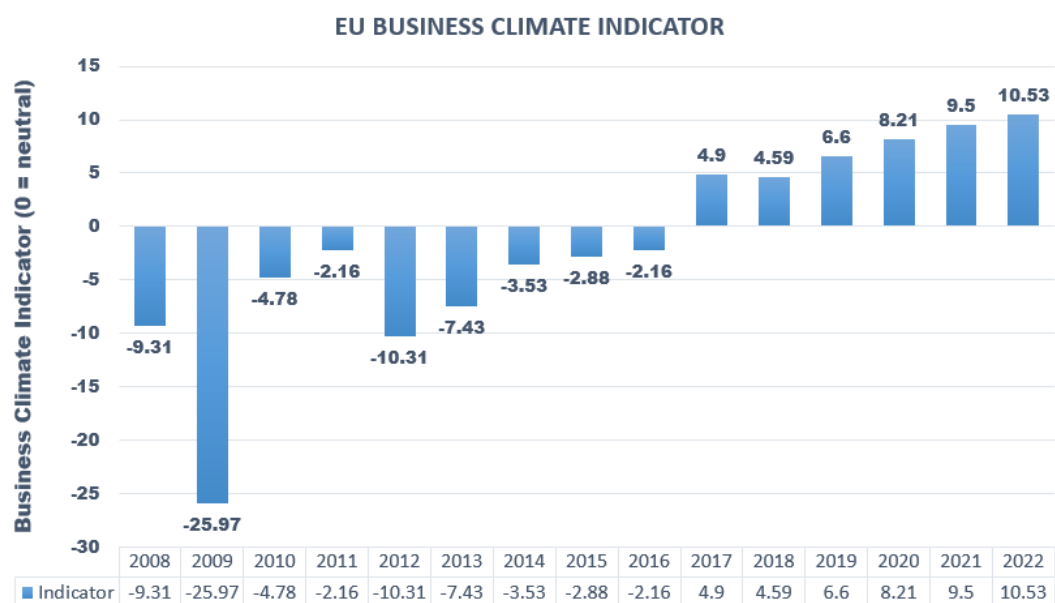


Fig. 2 Annual business confidence index for the industrial sector in the European Union from 2008 to 2022.
Source: Statista (2018)

The industrial sector plays a critical role in the Czech economy, accounting for approximately one-third of its GDP. With the sector's potential for growth, as evidenced by the European Union's confidence index, the Czech government recognizes the importance of fostering initiatives that bring together stakeholders to enhance the country's competitiveness in the industrial sector. In response to this, the Ministry of Industry and Trade (MIT) has spearheaded efforts to collaborate with policymakers, industry associations, businesses, and academia to establish a framework that supports the needs of the industries (Klitou et al., 2017). The government has implemented a comprehensive action plan to ensure that all stakeholders work in harmony towards achieving their goals, and the success of these initiatives will have a significant impact on the Czech economy's future. This highlights the government's commitment to leverage opportunities presented by the fourth industrial revolution while addressing the economic and social changes that come with it.

3.6 The current state of the Czech digital economy (Industry 4.0)

The fourth industrial revolution is a transformative global phenomenon that is rapidly reshaping the way we live and work, and is expected to have a significant impact on economies and industries worldwide. This revolution, characterized by the integration of advanced technologies such as artificial intelligence, the Internet of Things, and robotics, is fundamentally changing the way businesses and societies operate, and is creating new opportunities and challenges (Schwab, 2017).

As this revolution continues to unfold, it is becoming increasingly important for policymakers and industry leaders to understand the various situations and challenges that different economies and industries are facing in order to determine the extent to which they have absorbed this revolution. Such an assessment is crucial for identifying the necessary measures that need to be taken in order to improve the situation in areas that are lagging behind, as well as to capitalize on the benefits of this technological revolution.

At its core, collective assessment of the fourth industrial revolution should be made, that seeks to provide insight into the ways in which different economies and industries can adapt to the changes brought about by this revolution, and to identify the opportunities and challenges that lie ahead. By analysing the unique circumstances and challenges faced by different regions and sectors, policymakers and industry leaders can develop tailored strategies and policies that can help them navigate the complexities of this transformative era, and harness its full potential for growth and development.

In this chapter, the assessment will be viewed from the perspective of the state of the digital economy of the Czech Republic and how it is fairing against her fellow member countries in the European Union from the Industrial standpoint.

3.6.1 Digital Progress in the Czech Republic

The European Commission has been closely observing the progress of European countries in developing their digital economies every year. Through a comprehen-

sive set of quantitative measures, the Commission has been assessing various features of the economy in order to produce an annual report on each member country's performance since 2014. The ultimate goal of this initiative is to identify areas where each country needs to focus more attention and resources to improve their digital economy, and to provide guidance for the making and implementation of country-based policies. The report is measured using the Digital Economy and Society Index (DESI), which is a composite index that evaluates five general criteria: Human Capital, Use of Internet, Connectivity, Integration of digital technology, and Digital Public Services. This index is designed to provide a comprehensive evaluation of a country's overall digital economy performance and ranges from zero to one (0 – 1), with zero being the lowest, and the higher the number gets, and closer it gets to one, the better the performance of the country (Poor & Basl, 2018).

Here, the DESI criteria are being described according to the factors that constitute their measurement since its inception in 2014 (Adámek, 2018).

- The first criterion, **Human Capital**, has to do with the level of digital skills and education of the country's population. It includes indicators such as the percentage of individuals with basic digital skills, the number of graduates in science and technology, and the availability of training programs for digital skills development.
- The second criterion, **Use of Internet**, measures the level of digitalization of the economy and society in terms of the usage of digital services and platforms. This includes indicators such as the percentage of individuals using the internet, the frequency of e-commerce activities, and the adoption of online services by businesses.
- The third criterion, **Connectivity**, assesses the quality and coverage of high-speed internet access across the country. It includes indicators such as the availability of high-speed broadband infrastructure, the percentage of households with access to fast internet, and the extent of 4G mobile network coverage.

-
- The fourth criterion, **Integration of digital technology**, evaluates the extent to which digital technologies are used in different sectors of the economy, including manufacturing, healthcare, and transportation. It includes indicators such as the level of investment in digital technologies by businesses, the extent of adoption of cloud computing and big data analytics, and the deployment of digital solutions in public services.
 - Finally, the fifth criterion, **Digital Public Services**, measures the availability and quality of digital public services in areas such as e-government, e-health, and e-education. This includes indicators such as the level of online public service usage, the availability of online medical consultations, and the extent of e-learning opportunities.

However, according to the publication of the 2021 DESI report, the main criteria used were restructured to only four in, taking out the “Use of Internet” that had been included in the analysis for previous years, and adjusted to the exclusion of the UK from the European Union (DESI, 2021). Due to these changes, the DESI ranking for 2021 and 2022 that are up to date with the latest DESI ranking indicators, were considered in this review.

Tab. 1 Overall DESI ranking expressed in percentage

	DESI 2021	DESI 2022	Percentage growth
Rank	18	19	
Czechia	47.4	49.1	3.59
EU score	50.7	52.3	3.16

Source: DESI 2021 & DESI 2022, European Commission.

Table 1 displays the DESI score for the Czech Republic (Czechia) for 2021 and 2022, against the average score of the whole European Union (EU score) and how they rank (Rank) in relation to the other EU member states. In 2021, the Czech Republic scored 47.4% which was below the EU average of 50.7%, ranking the country 18th

among their European counterparts. However, the Czech DESI score grew to 49.1% in 2022 but still ranked lower than the previous year, finding themselves in a step lower (19th) than the year before as the EU average also increased to 52.3%. However one could argue against the worse ranking in a manner that, even as the European average experienced a percentage growth of 3.16%, the Czech digital economy experienced 3.59% percentage growth in its DESI score, from 2021 to 2022, being higher than the EU average growth in the score. This could imply that Czechia performed better than average in the EU in their digital economy growth. The 2020 DESI score and ranking across the EU is shown in Figure 3 below.

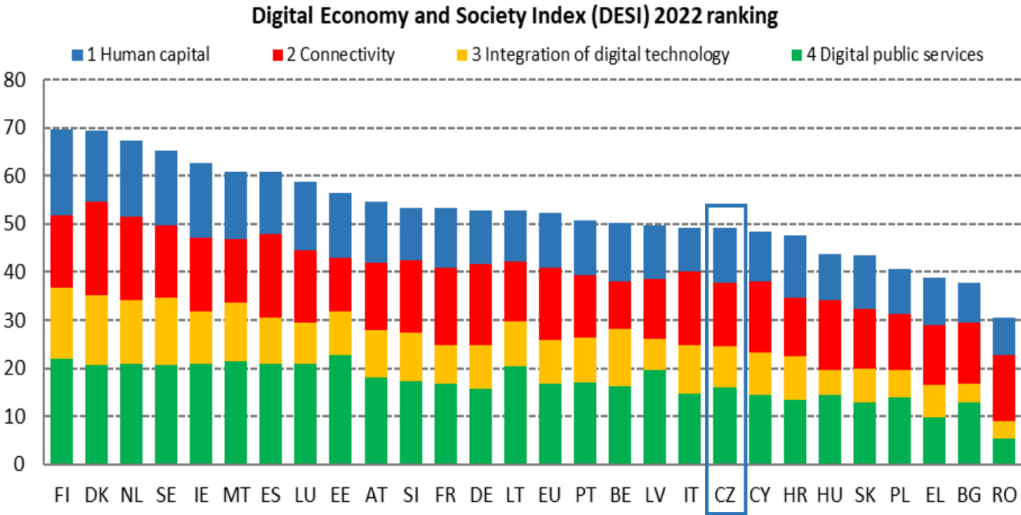


Fig. 3 Digital Economy and Society Index (DESI) 2022 for EU
Source: DESI 2022, European Commission.

Per Figure 3, the ranking is made up of the total scores of the respective countries from the four criteria considered for the digital economy and society index. With emphasis on the Czech Republic, Table 2 shows the DESI score by criteria (Score) for the years 2021 and 2022, how Czech ranked in each criteria (Rank), and the average criteria score of the EU (EU score). The contribution of each of these criteria towards the DESI index score for the years 2021 and 2022 are represented in Figure 4 and Figure 5 respectively.

Tab. 2 Czech DESI ranking by Criteria

	2021			2022		
Dimensions	Rank	Score	EU score	Rank	Score	EU score
Human Capital	15	47.2	47.1	15	45.6	45.7
Connectivity	22	44.6	50.2	17	52.7	59.9
Integration of digital technology	15	39.1	37.6	19	33.8	36.1
Digital public service	20	58.6	68.1	17	64.5	67.3

Source: DESI 2021 & DESI 2022, European Commission.

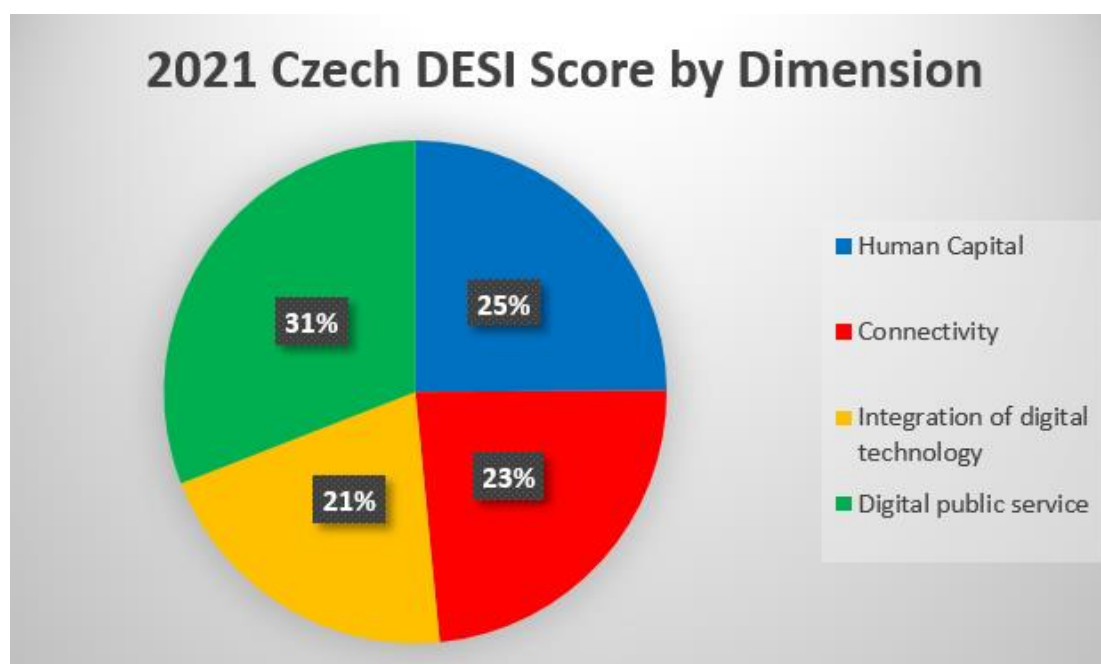


Fig. 4 Share of DESI criteria in Czech's total DESI score 2021
Source: DESI 2021, European Commission.

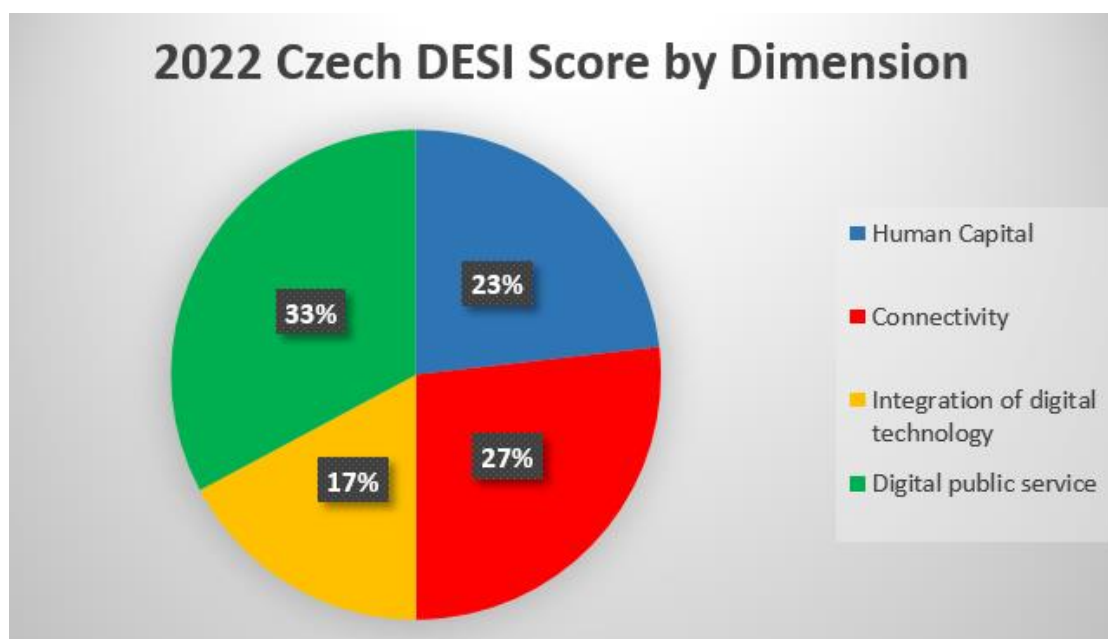


Fig. 5 Share of DESI criteria in Czech's total DESI score 2022
Source: DESI 2022, European Commission.

3.6.2 Implementation rate of Industry 4.0

A research study, “Assessing the industry 4.0 European divide through the country/industry dichotomy”, which was done to assess the implementation rate of the fourth industrial revolution across the European Union, assessed the countries under three broad categories (Castelo-Branco et al., 2023). This evaluation considers the various implementations and use of advanced technologies across different industrial sectors such as hospitality, construction, retail & wholesale, Logistics, amongst others according to classification of economic activities in the European community. Considering the cross-industrial evaluation, the researchers go on to look at the performance of these diverse industries in the implementation and use of Information and Communication Technology (ICT) in enterprises, as well as the major drivers of the fourth industrial revolution. The researchers then select the most viable and influential factors (constituent variables) across these countries and economic activities, and then groups these factors into major categories, that can easily assess the implementation rate of Industry 4.0 factors and at the same time, cover every essential factor that will be needed to determine the state of digital advancement. These categories are listed as implementation factors in Table 3 below (Castelo-Branco et al., 2023).

Tab. 3 Industry 4.0 implementation factors

Implementati on factor	I40 Infrastructure (Fundamental Business Operation)	Big Data Maturity	I40 Applications (Stages of industry 4.0 Implementation)
<i>Constituent variables</i>	• Interconnectiv ity. • Virtualization. • Collaboration. • Technology that connects business to customers.	• Big Data from any data source. • Data from geolocati on of portable devices. • Data from enterpris e's own smarts devices or sensors.	• Enterprise resource planning (ERP) systems. • 3D Printing. • Industrial/serv ice robots.

Source: Castelo-Branco et al., 2023

The researchers categorized these variables they used in their implementation rate findings with statistics of 28 countries including the United Kingdom (UK) which was still in the EU, from Eurostat database for the year 2018/2019. The factor score for some of the individual countries relevant to this study are shown in Table 4, with EU28 being the weighted average of all EU countries per industry considered for the analysis.

Tab. 4 Factor scores per country

Country	I40 Infrastructure	Big Data Maturity	I40 Applications
Austria	-0.08	-1	0.39
Czech Rep.	-0.32	-0.4	0.35
Denmark	1.34	-0.37	0.74
EU28	-0.24	0.06	0.06
Finland	1.31	0.64	0.87
France	-0.68	0.72	0.19
Germany	-0.49	0.33	-0.02
Greece	-0.79	0.25	-0.46
Hungary	-0.2	-0.67	-0.66
Italy	-0.67	-0.35	0.02
Luxembourg	-0.29	0.81	0.26
Netherlands	1.04	0.75	0.28
Slovakia	-0.13	-0.49	-0.31
Spain	-0.29	0.11	0.51
Sweden	1.29	-0.58	0.38
United Kingdom	0.29	-0.11	0.03

Source: Castelo-Branco et al., 2023

From Table 4, Czech Republic is observed to be among the lowest 30% of countries implementation rates with regards to the I40 infrastructure, with -0.32 factor score. The leaders in this category are seen to be Denmark, Sweden and Finland with 1.34, 1.29 and 1.34 respectively, while the bottom has big economies such as Germany (-0.49), France (-0.68), Italy (-0.67) and Greece (-0.79) all with poorer factor scores in I40 infrastructure than the Czech Republic.

Czech Republic scores even worse in the use of Big data, being in the lowest 25% with -0.40 factor score, nonetheless doing better than countries such as Austria

(-1), Sweden (-0.58) and Slovakia (-0.49). The countries with the top factor scores can be observed to be Luxembourg (0.81), Netherlands (0.75), France (0.72) and Finland (0.64).

In contrast, the Czech Republic is observed to be highly competitive in the I40 applications category, with 0.35 score points, placing them in the top 28% of countries together with Denmark (0.74), Spain (0.51) and Finland (0.87).

From the observations from this research, one could see that not one country dominates in all categories, with some countries scoring higher in some categories, and performing poorly in the other. As the Czech Republic falls behind its European counterparts in these driving factors of the fourth industrial revolution, it is an obvious call for actions to be taken, in order to improve the country's competitiveness amongst the European Union member states in the wake of the digitization revolution. Just as Jan Mládek, former Minister of Industry and Trade said, "Industry 4.0 is a way to maintain our competitiveness" (Klitou et al., 2017).

4 Research Methodology

4.1 Introduction

This chapter elaborates the target population and the sampling methods and processes that were used to obtain the population sample, as well as the source of data, research instruments used in data collection, and methods of data analysis. The research methodology is a crucial component of this work and it helped ensure that the research was conducted in a systematic and rigorous manner.

4.2 Study Design

The study design refers to the overall approach that was employed by the researcher to make the research practical, empirical, and address the research questions. In this study, a quantitative research design was used, which involved the collection and analysis of data through a structured descriptive survey from a select group of respondents (retail workers).

The questionnaire for the target group was made up of three major parts; PART 1 (Background Information), PART 2 (Readiness and Implementation of Industry 4.0) and PART 3 (Respondent's observations). The PART 2 of the questionnaire has two sections, whereby the first section assessed the current state of implementation of industry 4.0 in retail, and how the employees were equipped for these changes. The second section assessed how the employees see the future impacts of the industry 4.0 on their job security, skillset and probable demands that may come with technological changes in the retail environment.

In PART 3 of the questionnaire, the respondents have the chance to share their views on some open questions to enable them formulate their own opinions on the topic. A sample of the questionnaire can be found in [attachments](#).

4.3 Sources of data

This describes the means through which the data for this research was collected. This research employed the use of primary data to address the specific objectives of this research. The primary data used for this research was collected with a survey questionnaire. The questionnaire was administered and designed using a google online survey format, and the link was shared with various respondents. The questionnaire includes a statement of confidentiality and anonymity in the use of the response collected to ensure privacy protection as well as unbiased processing of this data.

The target respondents were mainly workers found in the various retail outlets in Brno and its surroundings. The level of education of respondents ranged from basic education to university education, between the ages of 15 – 65. Any worker in the retail, either part-time or full time, as well as persons who have worked for more than 3 months were considered for this survey, provided they were willing to take part in the study by providing accurate information to the researcher by responding the questions from the questionnaire which is added in [attachment](#).

4.4 Unit of analysis

Unit of analysis refers to the institution or population that are the focus of this research. In this paper, the unit was the various workers in the retail market in Brno, as this research aimed to collect data on how the fourth industrial revolution may impact the retail labor market. This ranged from, but not limited to, retailers in clothing and fashion, global food and supermarket chains, stationary and many more. The analysis was in the form of collecting their responses on the workers perception on the impacts of the fourth industrial revolution on the retail sector.

4.5 Population and Sample Frame

The population comprises the entire group of individuals who meet the criteria for inclusion in the study, thus retail labor in the Czech Republic. From this population, the researcher drew the sample frame, targeting retail employees within the South Moravia region of the country.

4.6 Sampling size

The sample size defines the number of participants selected to be in the study. In many cases, the population may be very large to allow for total inclusion in the study and in this case, the researcher carefully selects a befitting number as a sample to represent the population as they must represent the criteria of the sample frame it is obtained from.

In the selection of the sample size, the researcher used the Slovin's formula to help us make a considerable selection of size to make generalization of findings acceptable. The Slovin's formula is written as,

$$n = N / (1 + N e^2)$$

Where n refers to the sample size, N is the Population, and e is the error tolerance, derived from the confidence level the researcher intends to use. The researcher intends to attain an accuracy of at least 95%, hence a confidence level of 0.95, which means that an error tolerance or alpha level of $e = 0.05$ which is an acceptable optimal level for deriving a sample size (Tejada & Punzalan, 2012).

The population size we will use is the total number of employees corresponding to wholesale and retail trade; including repair or motor vehicles and motorcycles in the Czech Republic, from data obtained from the Czech Statistical Office. According to resources and dataset available upon thorough research, this category remained the most updated, closest and most accurate set of data needed for the purpose of this research, with the latest information corresponding to the year 2021 (Czech Statistical Office, 2022). In this case, $N = 577,026$ people. Per the Slovin's formula, the appropriate sample size will be 399.723, rounded up to 400 retail workers.

4.7 Sampling technique

In the writing of a research work, various sampling methods can be employed to select a sample from a population. These methods include random sampling, systematic sampling, convenience sampling, and purposive sampling. Systematic sampling involves selecting every n th element from a population after randomly selecting the starting point. Convenience sampling is a non-probability sampling method that involves selecting participants who are readily available or easily accessible. Purposive sampling involves selecting participants who meet certain criteria or characteristics that are important to the research question. However, these methods may introduce bias if the selection criteria are not truly representative of the population. As a result, random sampling is considered the most unbiased method since it provides an equal opportunity for every member of the population to be selected. The method is straightforward and easy to apply, which makes it possible to generalize the results obtained from the sample. The random sampling technique was utilized in the study to ensure that all individuals within the defined population had an equal and independent chance of being selected as part of the sample. The method is less complex, and it is easy to comprehend for researchers and other interested parties.

In this case, 400 retail workers were selected randomly to take part in this study.

4.8 Method of data collection

For the purpose of assessing the impacts of the fourth industrial on the retail environment with respect to the labor market, the researcher used questionnaires to collect data from randomly sampled retail employees.

4.8.1 Questionnaire

The questionnaire was distributed to the potential respondents in the Czech language since it is more comfortable and comprehensive for the respondents. The questionnaire in both English and Czech languages are added in [attachments](#) in this paper.

4.9 Method of Data analysis

The researcher used Excel and SPSS software tools to enter and interpret the data. The data collected via google forms questionnaire are represented in excel form to be interpreted. Excel was selected due to the fact that it was able to display the responses and adjust them correctly, and made them easier to understand and be used for regression analysis in SPSS analytical software.

Using Excel and SPSS, the researcher was able to analyse the data according to the sections the questionnaire was designed. From this it was possible to make derive descriptive relationships and statistics from the responses collected, in order to make meaningful interpretations on answering the research questions that need to be addressed. Various tables and charts that resulted from the questionnaire and data analysis will be presented in the next chapter.

5 Data Analysis and Interpretation

This chapter of the paper plays a crucial and pivotal part as it sheds light on the outcome of the dataset of the study. This chapter presents the results from the questionnaires that were administered, thereby bridging the gap between data collection and formulating meaningful insights to enable the researcher move closer to answering the research questions and make meaningful contributions to the existing body of knowledge aforementioned in the earlier chapters. The results of the data collection are then eventually presented in tables and charts that give a better insight on the viewpoint of the retail workers who were the main respondents to the questionnaires. The analysis covers the demographic and background information of the various respondents, their perspective on the current implementation of the fourth industrial revolution in their workplace, and their perspective on the impact new technology has in their workplace.

5.1 Demographics

It is imperative to the assessment of the data, to know the background of the respondents in the research. This serves as a representation and acknowledgement of the retail labour perspectives and account to forming conclusions that would be generally acceptable. Four hundred and six (406) retail workers have provided answers to the questionnaire, resulting in 101.5% response rate, considering the needed 399.7 (400) respondents that were required to make generally accepted conclusions. The dataset was however cleaned as some respondents left out essential information needed for full and meaningful assessment of their responses hence nullifying those responses. There were removed from the final dataset, leaving the researcher with three hundred and ninety-nine (399) final responses to be computed for the final analysis in this paper, which is still 100% of the estimated respondents that were initially required for this study.

Tab. 5 Gender of Respondents

Gender (Sex)	Frequency	Percentage
Male	67	16.8 %
Female	331	82.95 %
Non-Binary	1	0.25 %
Total	399	100 %

Source: Author's own (2023)

Table 5 above, highlights the gender of the respondents of the administered questionnaires. As presented, Sixty – seven (67) of the respondents were male employees in retail making 16.8% of the respondents' population. Three hundred and thirty – one (331) of the respondents are females, making 82.95% of the total respondents, and one (1) respondent identifying as non - binary, neither male nor female, making up to 0.25% of the respondents.

Tab. 6 Age of Respondents

Age (range in years)	Frequency	Percentage
15 - 19	0	0 %
20 - 30	199	49.9 %
31 - 45	122	30.6 %
46 - 60	78	19.5 %
60 and above	0	0 %
Total	399	100 %

Source: Author's own (2023)

Table 6 depicts a predominantly youthful labour force, but also a well-balanced and inclusive one. 49.9% of the respondents, making half of the entire sample for One hundred and ninety – nine (199), fall between the ages of 20 – 30 years. One

hundred and twenty – two (122) of the respondents, accounting for 30.6% of the respondents also fall in the category of 31 – 45 years of age. It could be said from these numbers work in the retail market is an attractive job to the youth, having 80.5% of the respondents between the youthful ages of 20 – 45 years. It may also be said that the nature of the job requires the energy of youthfulness in order to cope with the various tasks and that may be why there is a large number of young population in the sector, as compared to the seventy – eight (78) respondents who are between the category of 46 – 60 years of age, nonetheless a significant number as it makes up to a fifth of the respondents (19.5 %).

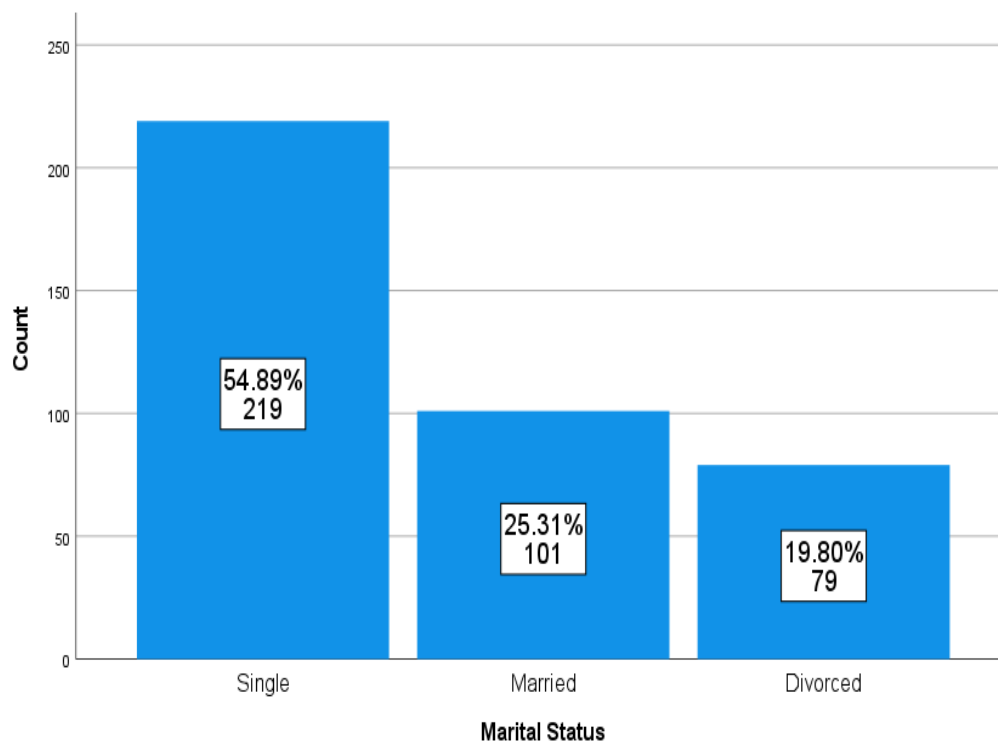


Fig. 6 Marital Status of respondents
Source: Author's Own (2023)

Figure 6 is a bar chart that shows the total number, and percentage of the respondents' marital status. Two hundred and nineteen (219) of the respondents are single,

One hundred and one (101) of them are married, and seventy-nine (79) of the respondents are divorced, having the respectful percentages of 54.89%, 25.31% and 19.8% respectively. There was an option for “other”, in the case where the status was not any of the three mentioned above, but none of the respondents were in this category. “Other” in this case could be a respondent that is widowed, separated, or any other status that may apply.

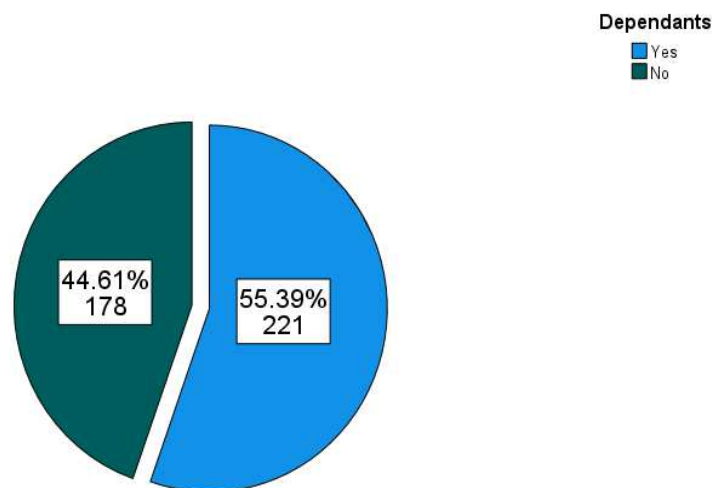


Fig. 7 Level of Dependency
Source: Author's own (2023)

The Pie chart above in figure 7 shows the level of dependency of the respondents. This refers to whether the respondents have kids, family members, or anyone they take care of, or any one that depends on them for a living. Two hundred and twenty – one (221) of the respondents answered they have dependants, that is 55.39% of the respondents. All other respondents answered they have no dependants, a total of One hundred and seventy – eight (178) retail workers, making up for 44.61% of respondents.

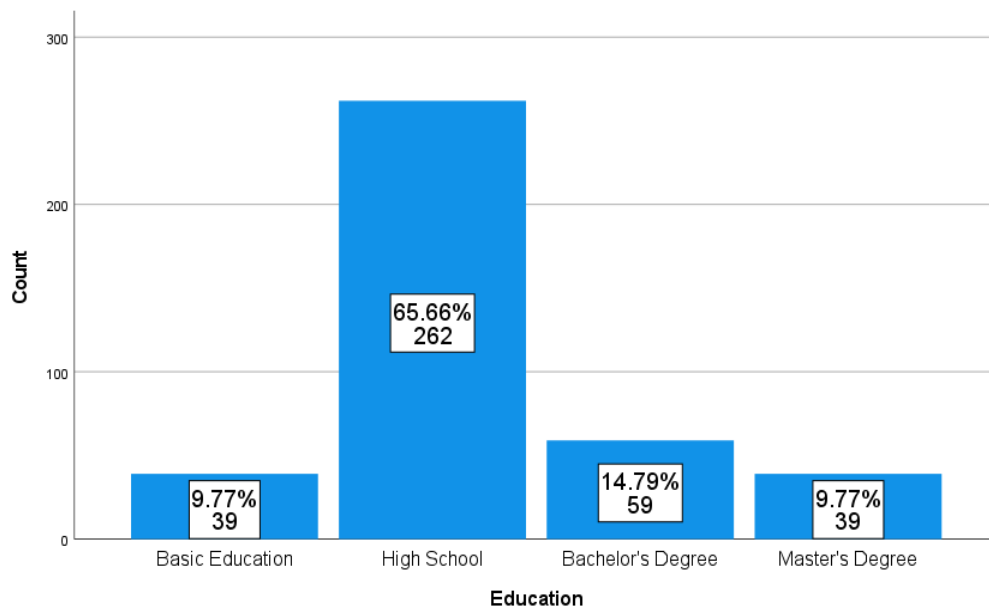


Fig. 8 Level of Education
Source: Author's own (2023)

From the figure 8 above, the highest form of educational level achieved by the retail workers that responded to the questionnaire are displayed. The original questionnaire had 5 levels of education to choose from, one of which are not represented by the bar chart above. This option was “Other form of education”, with particular reference to Technical or Vocational training. This has a frequency of zero, meaning that none of our respondents have this form of education as their highest level. Next in order are those retail workers with only Basic education. Thirty - nine (39) of the respondents selected this as their level of education, the same number as those with Master’s degree in this sector. Together, Basic education holders and Master’s degree holders make up to 19.54% of the respondents as each category recorded 9.77% each. Bachelor’s degree holders make up 14.79%, that is, Fifty - nine (59) respondents in the survey. The majority of respondents, two hundred and sixty - two (262) of the retail workers who answered the questionnaire selected that their highest level of achieved education was at the High school level, recording a whopping 65.66% of the total respondents. It is then evident that workers in this sector may not have a limit to the educational requirement, and even if there is, then it has

to be specifically position related. In other words, one do not need to have a high level of education in order to work in the retail sector, which gives it a higher probability of employing people with low to no education or technological skills. This will mean that every worker in this sector may need skills and on-the-job training irrespective of their educational background.

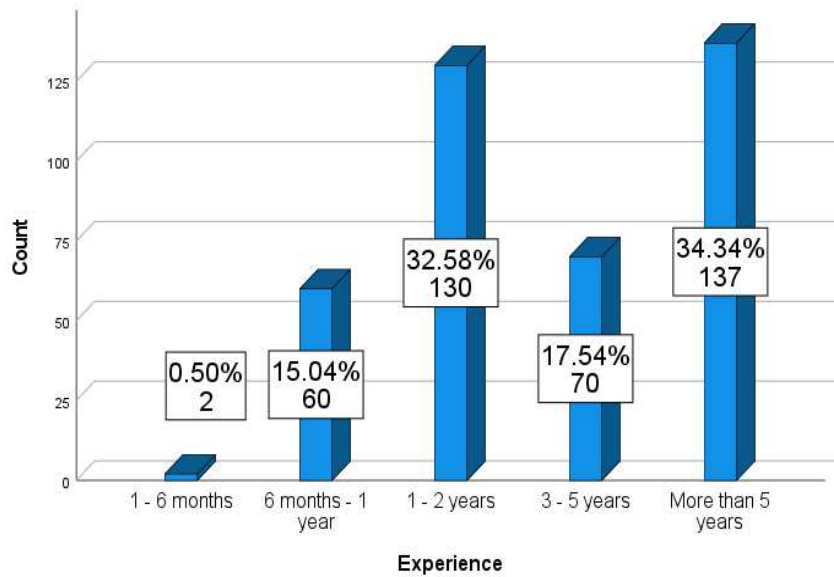


Fig. 9 Work experience (Months & Years)
Source: Author's own (2023)

The working experience of the respondents at the time of data collection have been displayed in figure 9 above. Only two (2) of the respondents have worked in the retail sector for six months or less, a mere 0.5% of respondents fall in this category. 15.04% of the respondents have been working in retail between six to twelve months (a year), making a total of sixty (60) respondents. One hundred and thirty (130) of the respondents, which make up 32.58% stated that they have been working in this sector between one to two years, Seventy (70) respondents (17.54%) have worked in retail for three to five years, and One hundred and thirty – seven (137) respondents have been working in the retail sector for more than 5 years, a

proportion of 34.34% of respondents. This shows that, approximately 84.46%, accounting for three hundred and thirty – seven (337) of the respondents to this questionnaire have worked in the retail sector for more than a year, where two hundred and seven (207) of them have more than three years' experience working in the retail sector. This level of working experience makes the views of such labour important and acceptable in measuring the impacts of the fourth industrial revolution on the sector, as they have worked in the sector over the years to go through various annual changes and developments that have occurred.

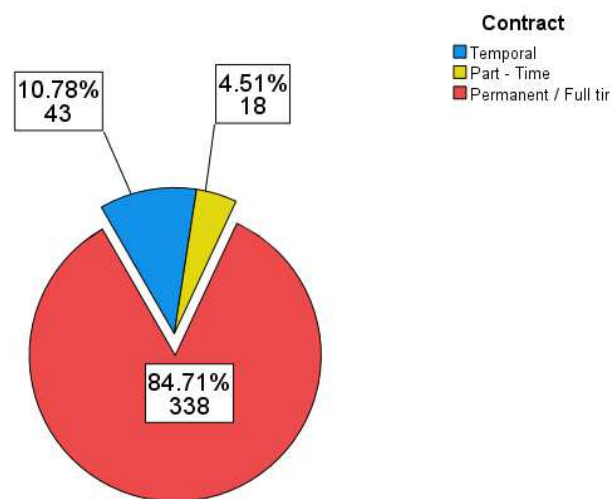


Fig. 10 Type of Employment Contract
Source: Author's own (2023)

Forty – three (43) of the respondents are working on Temporal (Brigada) basis, that is, 10.78% of the respondents. Eighteen (18) of the retail workers, making 4.51% of respondents have a Part-time contract. As shown in figure 10 above, the largest proportion of the respondents 84.71%, a total of three hundred and thirty-eight (338)

have full-time employment in the retail sector. This corresponds with the 84.4% of workers who have worked in the sector for more than three years, described in the above paragraph. Having this statistic implies that, only less than a fifth ($1/5$) of the entire retail respondents, do not work as full-time employees in the retail sector and of which also have less than 3 years of work experience in the retail sector.

From figure 11, it is seen that the total of two (2) employees who have worked 6 months or less, are all working on temporal (Brigada) basis. For everyone who have worked between six to twelve months, a total of sixty (60) employees; forty – one (41) of them, 68.33% also work on Brigada basis, eighteen (18) of them have a part-time contract (30%), and only one (1) of them have a Full-time employment in retail (1.67%). Everyone else that has worked for more than a year in the retail sector who responded to the questionnaires have Full – time employment, a total of three-hundred and thirty – seven (337) respondents.

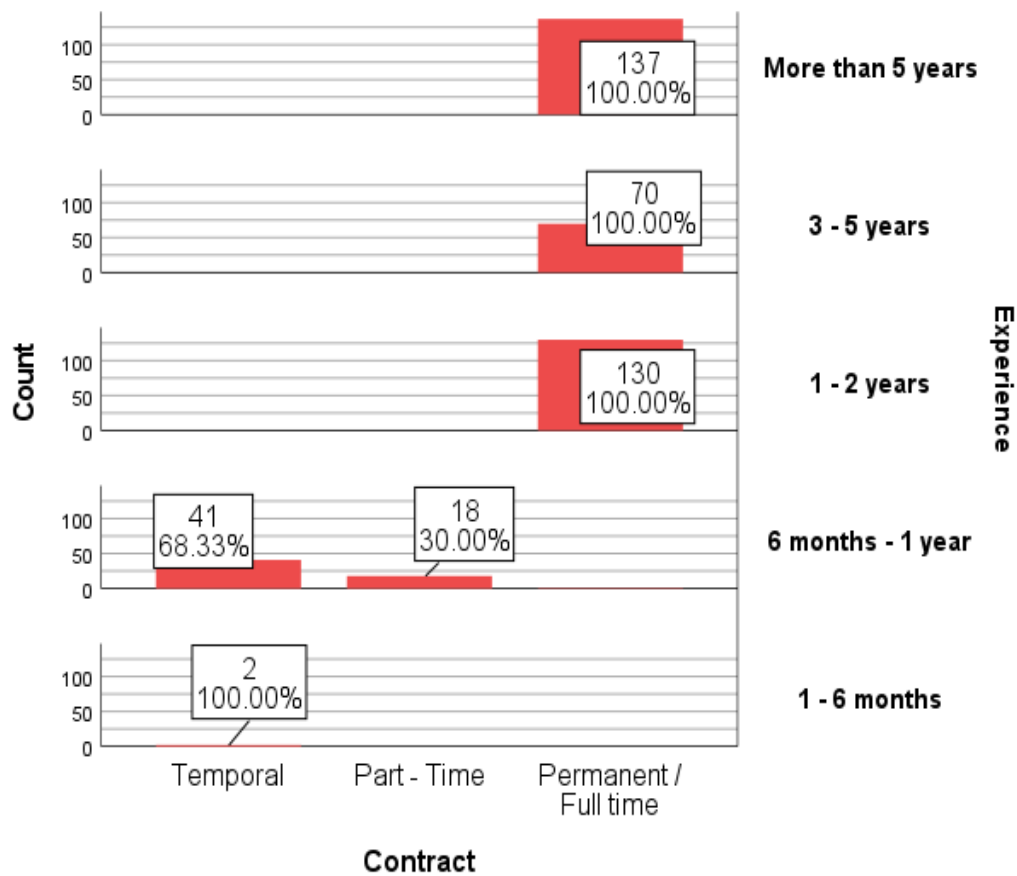


Fig. 11 Work Experience versus Contract Type
Source: Author's Own (2023)

Furthermore, the respondents were asked of their knowledge on the fourth industrial revolution. 5.3% of respondents which make a total of twenty – one (21) answered that they have well enough knowledge. The majority of Two hundred and ninety – eight (298) respondents (74.7%) answered that they know about the concept only a little, and eighty (80) respondents said they have no knowledge of the concept at all. This is represented in table 7 below.

Tab. 7 Knowledge on Fourth Industrial Revolution

Knowledge on Industry 4.0	Frequency	Percentage
Very Well	21	5.3 %
Just a little	298	74.7 %
Not at all	80	20 %

Source: Author's own (2023)

5.2 Reliability Analysis

Reliability, in the sense of measurement is a pivotal aspect of the research and assessment process that underscores the consistency and precision. In the sense of testing, it could be said to be any variation in scores that emerge as a consequence of factors that have been included in the measurement process but may be unrelated to the phenomenon that is being measured. Reliability analysis, put together in the sense of the above mentioned criteria can be said to be a defining attribute that embodies the degree to which scores of a particular assessment exhibit consistent and stable pattern and unadulterated by error (Amirrudin et al., 2021). It measures the ratio of the real score of variance to the total score of variance, drive for consistency and reveals measurement error for mitigation in order to establish a dependable instrument.

It could be said that whether or not a particular instrument is reliable, it does not depend on its validity. However, an instrument cannot be valid unless it is reliable hence a researcher is always advised to employ statistical methods to estimate the reliability of the measurement in question. Some of the types of reliability are Test – Retest Reliability which employs test-retest correlation by assessing the correlations on two different occasions. Another is Inter-Rater Reliability, which employ Kappa Statistic to assess categorical data, and also Internal Consistency Reliability which employs Cronbach's Alpha to measure how well items in a scale of a questionnaire correlate (Dunn, 1992). For the purpose of this research, the author employed the use of Cronbach's alpha to measure the reliability of scale used in the questionnaire. A Cronbach alpha below 6.0 is said to have low internal consistency and thus, should be subject to review so as to improve them, or revisit the questionnaire design. An Alpha between 6.0 and 7.0 can be considered to have moderate consistency which may not be good enough for a viable conclusion. A Cronbach alpha with 7.0 – 8.0 is considered acceptable with reasonable level of internal consistency, and any alpha above 8.0 is considered to be good and have a high level of internal consistency.

Tab. 8 Reliability Analysis

Variables	Cronbach's Alpha	Number of items
Implementation of technological Changes	0.74	5
Impact perception of Employees	0.76	7

Source: Statistical Package for the Social Sciences (SPSS)

Table 8 above depicts the variables that correspond to the classification of questions that were employed in the questionnaires that were administered to the retail employees. The first part, which had five (5) questions relates to the employees' perspective on how changes are implemented in their current workplace. The Cronbach's alpha for these questions are 0.74, which makes it an acceptable measurement scale for the implementation part. The second variable had eight (8) questions to assess how the employees agree with the impacts of fourth industrial revolution in their workplace, based on their experience with the implementation they have had at the time the data was collected. Upon further analysis, one of the questions were not a good measure for the impact measure and had to be taken out. Reason being, one of the appropriate items asked "*Do you think that new technologies will reduce the number of jobs in retail?*" while the item that was removed assessed the literal opposite of this question "*Do you think new technologies will create more/new jobs in retail?*", and hence it makes sense that it doesn't comply with the scale of measurement in this case. It has a negative relationship with this question, together with all the other questions in this category and hence had to be removed from all further analysis to avoid bias and inconsistency of the results of the analysis. This change resulted in a change of the Cronbach's alpha from 0.6 to 0.76, making the measure a good enough measure acceptable for assessing the customers perception of these impacts. The reliability analysis was performed with Statistical Package for the Social Sciences (SPSS) with the responses from the data obtained from the questionnaire.

5.3 Implementation of Technological Changes

The research findings through the questionnaires administered in this section asked employees some questions about the use of new technologies in their workplace and their implementation. The findings from this set of questions have been presented below.

Tab. 9 Implementation Conditions

Statement	Strongly Disagree	Slightly Disagree	Neutral	Slightly Agree	Strongly Agree
Your workload has changed (increased / decreased) due to new technology/ application/equipment	2	30	81	133	153
You get informed of new changes/improvements in your work before it happens	4	12	58	192	133
Management prepares you with training for any new changes before they happen	5	33	26	175	160
You have enough skills to cope with technological changes	0	21	203	119	56
You are able to adapt well to technological changes	9	20	89	185	96

Source: Author's Own (2023)

When the retail employees were asked on the various implementation questions, they provided positive responses in the majority of cases. With regards to whether the implementation of new technology has changed their workload, 71.6% of the employees responded with affirmation that indeed new technology has impacted their workload. 20.3% of them could not agree or disagree with this impact, while only 8% of respondents answered that their workload has not changed at all. It should also be noted that 81.4% of the employees agreed to the statement that they are informed about any new changes that would be introduced to them before it happens. 4% of respondents said they are not informed of new changes while 14.5% of the respondents had a neutral view on this statement. When employees have the heads-up on what changes are about to be implemented, it makes them feel involved in the decision making and helps them to orient themselves in a position to adjust to the changes that are about to happen. Every employee would like to feel included and readily prepared so as to not feel outcast or not important in their workplace, so there is no surprise that as a high percentage of employees responded positively to being informed of changes before they happen, a correspondently high percentage of them agreed that they get adequately prepared for the changes as well.

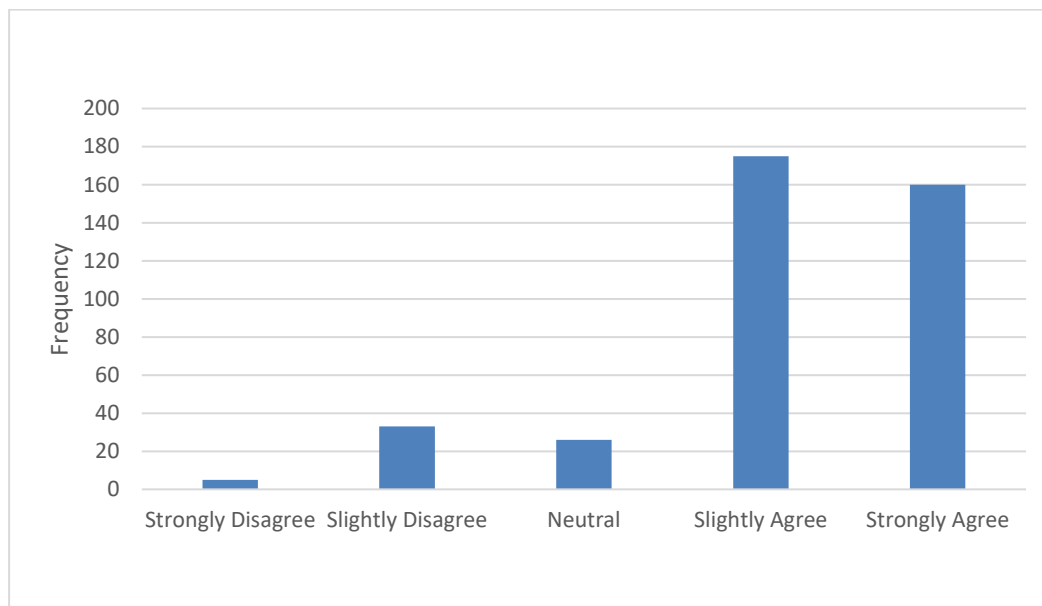


Fig. 12 Management prepares employees for new changes.
Likert Scale on management preparing employees for new changes, ranging from (1) Strongly Disagree, (2) Slightly Disagree, (3) Neutral, (4) Slightly Agree and (5) Strongly Agree.
 Source: Author's own (2023)

Figure 12 shows how much employees agreed with the fact that management gave them the necessary preparation for new changes before they happened. 84% of the respondents agreed with the statement that they get the needed training for new changes in technology that management plans to implement, before the changes actually happened.

It is therefore deemed imperative that employees get trained in order to gain the skills that will help them stay in their work and be more productive. Only 43.8% of the employees answered positively that they have enough skills to cope with technological changes. This proportion is even lesser than the proportion of respondents who have more than three years' experience (52%) in the retail sector. 50.9% of the respondents cannot agree or disagree with the fact that they have enough skills to cope with technological changes, and 5.3% of the respondents disagreed, implying they do not have enough skills to cope with technological changes. The high neutral-

ity in response seems valid to some extent, in the sense that they have always received training and preparation for new changes that happen as mentioned in the earlier paragraphs, hence there is a high possibility that they will require preparation with new skills that may be needed for new technology that may be introduced in the work place as the fourth industrial revolution is still on-going, and new innovations are being created constantly. This is represented in the figure 13 below.

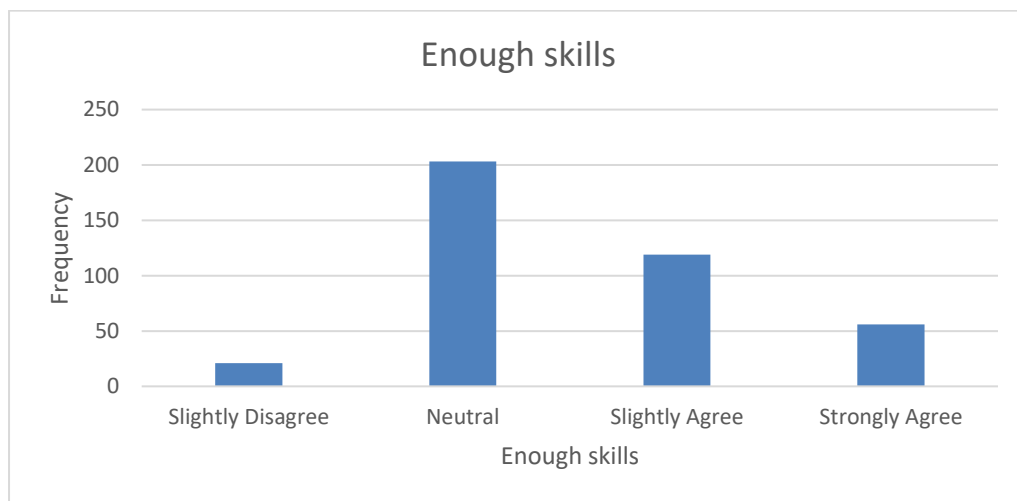


Fig. 13 Employees have enough skills to cope with technology changes
Likert Scale on current skillset of employees to cope with new changes, ranging from (1) Strongly Disagree, (2) Slightly Disagree, (3) Neutral, (4) Slightly Agree and (5) Strongly Agree
 Source: Author's own (2023)

For trainings and preparations to be smoothly executed and implemented, one factor it may boil down to is how well the employees are able to adapt to new technology and skills that may be needed to use them. Employees that took part in the questionnaire were not uniformly confident in their adaptability to new technologies. 70.5% of respondents are confident they are able to adapt well to new technologies, and only 7.3% of them responded that they are not able to adapt well to technological changes. A significant proportion of 22.3% cannot also agree or disagree that they can adapt well to changes that may come with new technologies. These proportions are illustrated in the pie chart in figure 14 below.

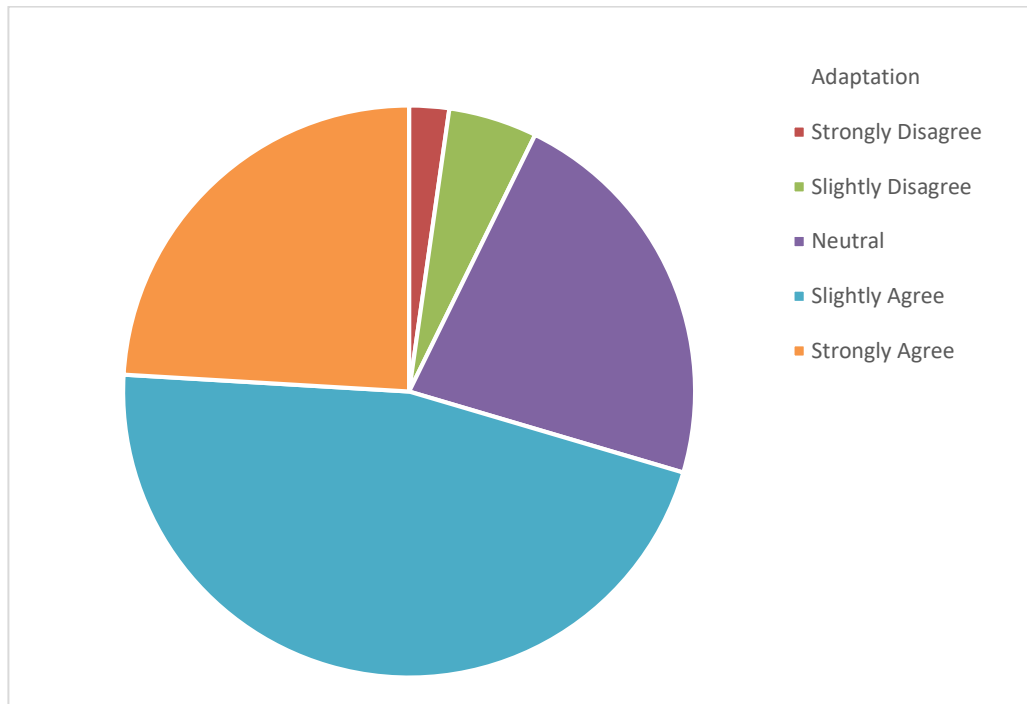


Fig. 14 Employees are able to adapt well to technological changes.
Likert Scale on Employee Adaptability to technological changes, ranging from (1) Strongly Disagree, (2) Slightly Disagree, (3) Neutral, (4) Slightly Agree and (5) Strongly Agree
Source: Author's own (2023)

The mean value for the responses in this 5 - item implementation variable discussed above was 3.92 with a standard deviation of 0.62. The components of the variable were measured using a five (point) Likert scale from the lowest being (1) strongly disagree to (5) Strongly Agree, being the highest. The standard deviation which tells how much a response's data point are from the average gives us insight on whether the variation in answers signify a strong or poor coherence. With a standard deviation of 0.62, the individual responses are not very far from the mean value, hence make the score of our data a good one. These descriptive statistics are represented in the table 10 below.

Tab. 10 Descriptive statistics

Variable	Mean	Standard Deviation	Number of Items
Implementation	3.92	0.62	5

Source: Statistical Package for the Social Sciences (SPSS)

5.4 Impact Perception of Employees

In this part of the questionnaire, the respondents were asked a few questions to access how they feel the impacts of the fourth industrial revolution in their work, based on the experience they have obtained from the implementation that has already taken place.

Tab. 11 Impact perception

Statement	Strongly Disagree	Slightly Disagree	Neutral	Slightly Agree	Strongly Agree
There will be lesser jobs in retail	0	30	27	182	160
Work-life balance will be improved	1	20	205	123	50
Means to perform work will change	0	10	58	198	133
Job security will be in threat	9	26	88	185	91
New skills will be needed	0	28	81	140	150
Your job will get harder to do	2	28	255	96	18
You will have less tasks to work on	1	5	126	178	89

Source: Author's own (2023)

The respondents were asked to select the appropriate degree of which they agree or disagree with impacts of the implementation of industry 4.0 on a 5 – point Likert scale ranging from 1 – 5, with the lowest as (1) strongly disagree to (5) Strongly Agree, the highest.

When the retail workers took part in this research, 85.7% of them responded in agreement to the fourth industrial revolution leading to lesser jobs in the retail sector. 40.1% strongly agreed with this statement, and the other 45.6% slightly agreed with it. This is not a surprising take as around 70% of the respondents answered

that the technologies that have already been implemented has affected the workload they currently have in the workplace.

Employees responded positively as to if they expect the changes in technology to significantly change the way their job is done. 82.9% of the agree to the significant changes the fourth industrial revolution will bring to how their job is currently performed, with 33.3% strongly agreeing and 49.6% slightly agree. Just as illustrated in the literature review chapter on how much basic work and everyday life has evolved from the first to second to third industrial revolutions, it is no surprise that the current retail workers expect these significant changes. The direction of these changes are however contentious as to whether it will make the retail job harder or much easier.

One of the elements in this section implored respondents to answer if the new technology will make their job harder than it is, and the employees' responses are displayed in Figure 15 below.

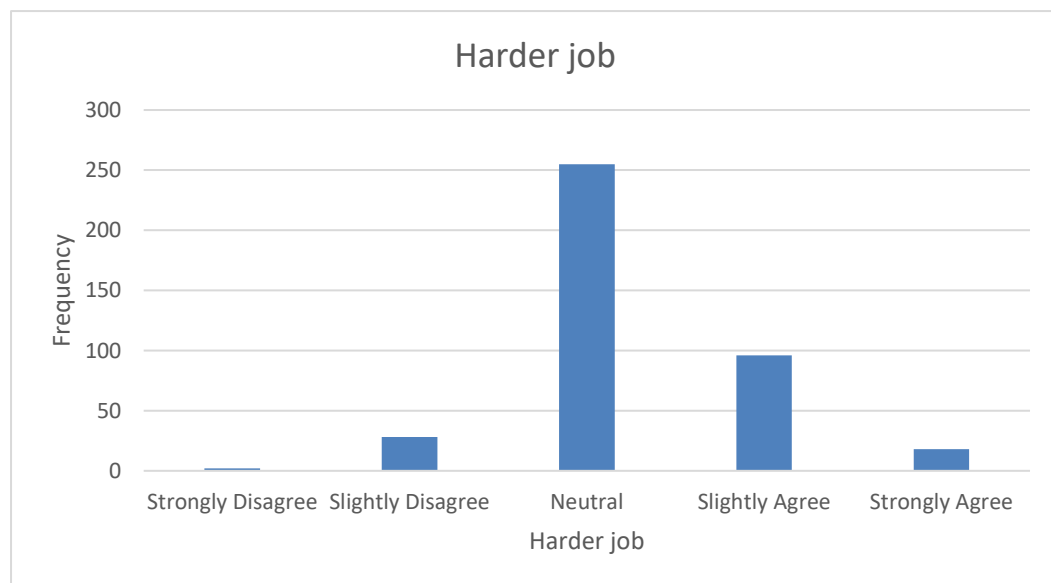


Fig. 15 New technology will make working in retail harder
Likert Scale on new technology making retail jobs harder, ranging from (1) Strongly Disagree, (2) Slightly Disagree, (3) Neutral, (4) Slightly Agree and (5) Strongly Agree
Source: Author's own (2023)

A higher percentage (63.9%) of the respondents answered they can neither agree nor disagree with the fact that new technology will make their work harder. When new technology is introduced, it may be hard for employees to work as they normally do since they need to be trained in order to get conversant with the new technology. 28.6% of the retail labor respondents however agreed that it will make their jobs harder, while 7.5% disagree and think it will make their jobs rather easier, but whichever way this is looked at, change in their working conditions are eminent, as they have confirmed in the section about change in the way their work is performed.

If indeed, these changes start to reduce the jobs in the retail sector the livelihood of these employees will be at risk as they will need to live without jobs, or try to find new jobs, which might end up affecting the work-family life balance of these employees. This reflection was evident in the way these retail employees responded to this kind of impact, with clear division in opinions, as the revolution is a slow process, even as it is spreading fast in every sector of the economy. Only 43.3%, less than half of the respondents agree that the fourth industrial revolution will improve their work-life balance, while 51.4% a majority of respondents could not agree or disagree that their life balance would be improved. It could be due to the fact that they do not relate the new technology to their family life, or the full effects of the industrial revolution cannot be currently measured in the retail sector, as the changes are still ongoing, understandably.

Whichever way you look at it, it is clear that the fourth industrial revolution will change the way retail work is performed as has been evident in many other sectors. If machines take over some of these manual jobs at the moment, they can only get better and hence will put the livelihood of many retail workers at risk. To affirm this conclusion, 69.2% of the respondents agreed that the new technology will pose a threat to their job security in retail. The bar chart in the figure 16 below, displays the degree to which retail employees agree or disagree that their job security is at risk with the implementation of Industry 4.0.

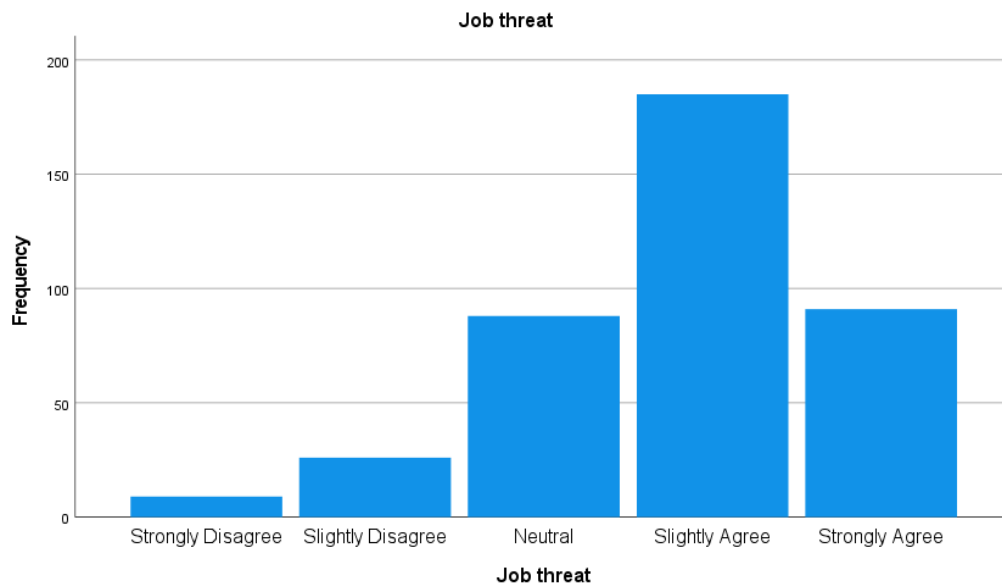


Fig. 16 New technology will pose threat to job security
Likert Scale on technology posing threat to retail job security, ranging from (1) Strongly Disagree, (2) Slightly Disagree, (3) Neutral, (4) Slightly Agree and (5) Strongly Agree
 Source: Author's own (2023)

In the face of these changes, it is either employees learn the skillset that are needed to work with the new technology or lose their job to the technology in case they do not possess the right skills to keep them working in the retail sector. The respondents were positive in their responds that they will require new skills and knowledge to work with the new technology. Out of the 72.7% of employees that agree that they will need new skills and knowledge, 37.6% of them strongly agree and 35.1% of them slightly agree. None of the employees responded that they strongly disagreed with this, but only 7% of them slightly disagreed, with 20.3% being in a neutral position. For the changes that are going on in the retail market, if an employee is confident with the skills they possess, they would disagree that they need new skills, and if one feels they need more skills to cope, then they are aware of their deficiency in working with the new technology. It then makes sense that respondents who are neutral may not know what skills they need, or what is yet to come which will determine the skills that may be required from them.

One thing that they are certain of though, is that using new technologies will make take some tasks away, as that is the whole idea behind digitization and employing the use of modern machines. From factories to farms to medicine to every aspect of human life and economy technology has always been employed to make work easier, faster and more efficient than it would have been with human in charge. It therefore makes sense that only 1.6% of the total respondents disagreed that the fourth industrial revolution will give them less tasks to work on. 66.9% of them in fact agreed that they will have lesser tasks to work on, which in implication speak that work will be easier, but there will resultantly be a lesser need for human in such positions.

The components of the variable were measured using a five (point) Likert scale from the lowest being (1) strongly disagree to (5) Strongly Agree, being the highest. The mean value for the responses in this 5 - item implementation variable discussed above was 3.83 with a standard deviation of 0.53. The standard deviation which tells how much a response's data point are from the average, where large standard deviation means it is poor and smaller standard deviation implies that the figures are much closer to the mean and are good. With a standard deviation of 0.53, the individual responses are not very far from the mean value, hence make the score of our data a good one. These descriptive statistics are represented in the table 12 below.

Tab. 12 Descriptive statistics

Variable	Mean	Standard Deviation	Number of Items
Industry 4.0 Impact perception	3.83	0.53	7

Source: Statistical Package for the Social Sciences (SPSS)

5.5 Open – Ended Responses

In this chapter the author delves into the intricate process of analysing open-ended responses from the questionnaire.

A variety of question types are often employed by researchers in the pursuit of comprehensive data collection in surveys, and while close-ended questions and Likert-scales were employed to obtain structure and easily quantifiable data to ensure smoother analysis, the researcher of this paper also employed open-ended questions to obtain a distinct dimension of nuance to the research process. Even though the closed – ended questions offered respondents a set of predetermined response options to express themselves, it was important to invite participants to share their thoughts and experiences in their own words to shed more light and give enough context, richness and deeper understanding of the impacts and implementation of the fourth industrial revolution in the retail sector.

The respondents were asked four open-ended questions as displayed in the first column of the table 13 below, with the corresponding number of responses in the second column for each question.

Tab. 13 Open-ended questions

Question	Number of responses
Can you share some tasks that were done by humans in your workplace that is now done with technology?	279
What changes do you expect in the future of retail work, considering the ongoing technological advancements?	228
What support or training, do you think will be beneficial for retail workers, to survive the changes brought by new technologies?	171
Is there anything else you would like to share on how the introduction of new technologies may impact working in retail?	255

Source: Authors Own (2023)

In this section of the questionnaire, the respondents had the chance to link their responses in the current implementation of new technologies, and their respective impact perception with their own words. These responses served as pillar of support for the numerical analysis that were explained in the chapters prior.

- *Can you share some tasks that were done by humans in your workplace that is now done with technology?*

When the respondents were asked to share some tasks that were manually done in their workplace that is now performed by technology, an array of answers were given but the ideas these answers were talking about were quite similar. Out of the two hundred and seventy – nine (279) responses received on this question, only 82% of them were valid responses which corresponds to two hundred and twenty – nine (229), and even those valid responses had only fifty – seven (57) of them (25%) talking about technological advances in other sectors. These were mostly about the use of robots in restaurants and hotels, the changes in agriculture like milking of cattle, digging and harvesting of vegetables and so on. However, the most relevant of these responses regarding retail had to do with online shopping, electronic price tags, network communication, invoicing and self-service.

29% of these respondents mentioned that pricing of goods, filling of forms, ordering stock and even taking inventory were done by hand and going around, but now, computer takes stock of inventory as when business is ongoing, and that computers automatically issue invoices to customers without having to write by hand. 33% of the respondents spoke about the use of self-checkout and hence no need for many cashiers, or people placing their orders online and picking them up or using delivery service without having to talk to employees to do these things for them. 13% of them also talked about the fact that communication with management, headquarters, suppliers and colleagues easier for them no matter whether they are, either at work, or outside work using social networks, and other internal infrastructures that has made communication and reporting faster and more efficient in the retail sector.

- *What changes do you expect in the future of retail work, considering the ongoing technological advancements?*

The respondents were also asked to share their views on what changes they expect to experience in future considering the ongoing technological advancement. Out of two hundred and twenty – eight (228) responses received, only one hundred and seventy – seven (177) were valid answers that can be examined. Out of this total, 41% of the employees responded to the fact that they expect there to be lesser and lesser employees in the shops. They explained that there could be more self-check-out panels and more technology incorporated into the shops and hence there will be less cashiers needed in the retail sector. 18% of the respondents to this question stated that they expect improvement in the current technology. Some of the peculiar views shared mentioned that they expected baskets that scan the items automatically when they are put in them, some express the fact that some items still need to be weighed or counted manually which takes time. The other 41% of the respondents expressed that they expect the use of employees to be obsolete in the retail sector. They attributed this to the use of automatic and electronic vacuums that will eliminate the need for cleaners in the shops, as well as electronic information boards that can give people the needed information which is currently pushing information desk attendants out of job and that there is the expectation to have more of these systems. Some of them also said that there the shops are going to operate only online and through applications just as banks have started to do, while others expect shops without people and attendants but only self-checkouts. Some even pointed out the fact that there are currently delivery boxes where one can just go and pick their items without the need for post man or delivery driver or have contact with them. With regards to this, the respondents expect that there will be the reduced, or no need at all for these attendants and fronts in the retail market in the near future.

- *What support or training, do you think will be beneficial for retail workers, to survive the changes brought by new technologies?*

The retail employees were also asked to share what kind of support/training they will need to survive the changes the new technology brings and their responses have been aggregated as follows. One hundred and seventy – one (171) responses were received for this question and only One hundred and fifty – one (151) of them were

valid for analysis. 39% of the valid responses mentioned that all they wanted was support from management and supervisors when they encounter difficult situations. Some of them mentioned that the most important thing is having the experience and support from co-workers and not any special training. On the other hand, 20% of the respondents answered that all they needed was training from a qualified instructor. They mentioned that training with a coach who would enlighten them on the use of technology will be the best to help them cope with the technological changes. Not only that, as some of the respondents in this category suggested that these coaches should be able to explain to the retail employees that technology is actually a good thing and it will make their work and lives easier. The remaining 40% believe in continuous training. Responses from this group mentioned alternatives like having e-learning and information available to the workers all the time so that they can stay up to date, being trained about new product and technology bit by bit as they come, since they claimed that the technological requirement in the retail field is not very difficult and with the basic knowledge of basic technology is all they need to keep up with the technological changes.

- *Is there anything else you would like to share on how the introduction of new technologies may impact working in retail?*

The last one of these open-ended questions that was posed to the retail workers was if there was anything else they would like to share on how new technologies may impact working in retail. This was meant to probe the employees to share any thoughts the respondents had in case they have not been mentioned early on and received two hundred and fifty – five responses (255). 70% of these responses were not valid to be used, as they all stated that these respondents had nothing to say on this question. With the few that actually provided some answers, 11% said that technology will harm them, another 10% said that they would actually embrace all changes if it makes their jobs easier, and the last 9% said that technology makes life easier and it will definitely be of benefit to the retail sector. Even though these are not enough responses to make any conclusions, the 30% of valid responses can be used

to make statements concerning the sample that took place in this survey as this accounts for seventy – seven (77) respondents.

There is no doubt that the implementation of Industry 4.0 has impact on the labour force in the specific sector, either negatively, positively, or both. This theory has been affirmed in the Chapter 3.4 of this paper, which reviewed existing literature on this Implementation – Impact relationship. On this basis, the author of this paper undertook this research to identify the impacts the industry can have on the retail sector labour but collecting data on implementation, and the impact perceptions the retail labour has on industry 4.0. In the chapter that follows, the author conducted correlation analysis on this data collected in order to back this relationship according to the data obtained from the survey employed for the data analysis.

5.6 Correlation Analysis on the Implementation and Impacts of Industry 4.0 based on questionnaire survey

The term correlation could be described as a relation existing between phenomena, things or statistical variables which tend to vary, be associated, or occur together in a way not expected by chance alone (Akoglu, 2018). The author employed the use of bivariate correlation in order to establish the relationship between the two variables measured in the data analysis of this research, being Implementation of the fourth industrial revolution, and its impacts as expressed by the respondents.

Since the data available or ordinal in nature, the total mean of the variables in question were calculated to make a continuous variable, in order for correlation relationship to be measured using the Pearson's r .

The author discovered that there is a significant positive relationship between the implementation variable and the impact variable, $r(397) = 0.91, p < .001$, which signifies a high positive correlation between the implementation of fourth industrial revolution and the impacts that it bring to the retail sectors involved in this research. This relationship is represented by the scatterplot diagram in figure below.

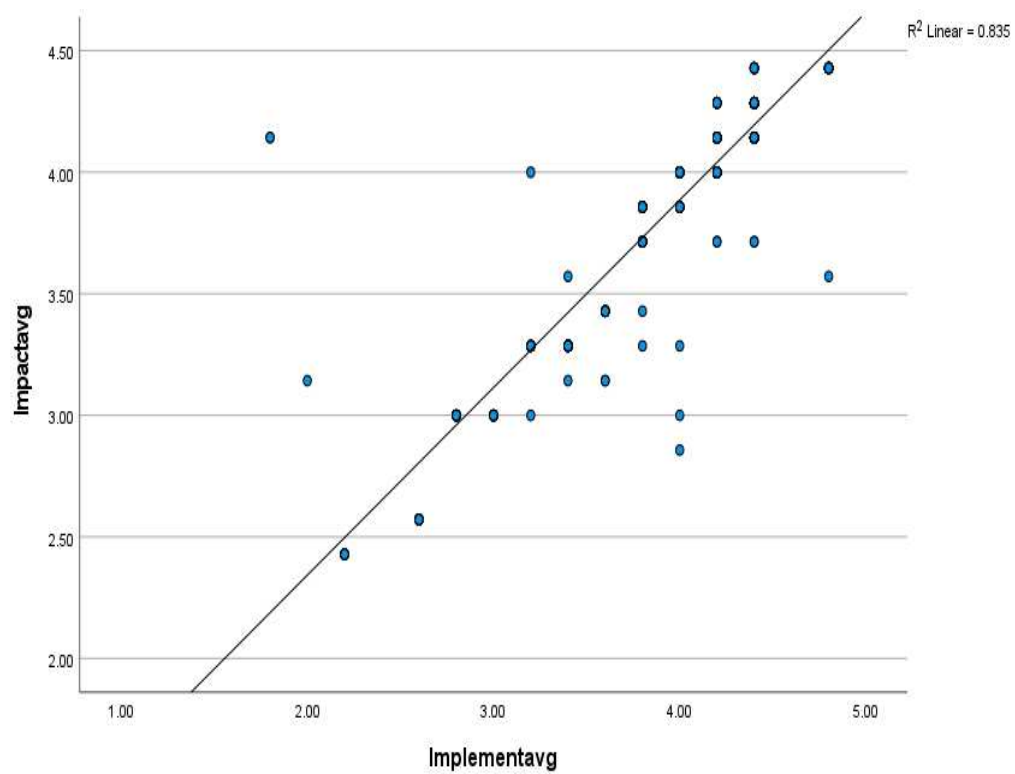


Fig. 17 Scatterplot for correlation between the variable means.
Source: Statistical Package for the Social Sciences (SPSS)

6 Summary of Results, Conclusions and Recommendations

6.1 Introduction

The main aim of this research was to assess the impacts of the fourth industrial revolution on retail labour in Czechia, Brno, in an attempt to propose recommendations that could be used to manage these impacts on the workers. In order to accomplish this, the research questions need to be answered. Firstly, how will these technological advancements affect the jobs and the retail labour market? Secondly, to what extent will the evolution impact the future of this industry? And finally why should the retail labor market embrace the evolution? This final chapter draws conclusions and makes discussions towards answering these questions, and attempt to propose useful recommendations to manage these impacts.

6.2 Discussion and Summary of Results from Data Analysis

6.2.1 Summary and discussion on the implementation of Industry 4.0 in retail

In this paper, the research survey which was conducted with the questionnaires gathered that fourth industrial revolution-related implementation have occurred in the retail sector. The Digital Economy and Society Index (DESI) which is used by the European Commission to observe the technological progress of its member countries are based on Human capital (level of digital skills of the population), Use of internet, connectivity, integration of digital technology, and Digital Public services (Adámek, 2018). For the purpose of this research, the most relevant of these factors to the retail sector include human capital, connectivity and integration of digital technology of which the employees in this research were asked to share their views on. On the basis of these implementation factors, an average of 71% of respondents Strongly/slightly agree that the new technologies have been introduced to their workplace and they have had to adapt. This is drawn from their responses to the

question of whether new technologies have changed their workload, where 71% of them agreed, and 70% of them affirmed that they were able to adjust well to the new technologies that are introduced in their workplace. On the implementation practice of management in the retail sector, 80% of the retail employees agree that their managements informs them ahead of technological changes that may occur, and give them the necessary training to cope with the changes that come with the new technology implementations, which also contributes towards an improvement to the human capital in the retail sector.

However, when the employees were asked if they have enough skills to cope with the technological changes, 51% could not agree or disagree whether they have enough skills, with only 43% of respondents affirming that they have enough skills to cope with these changes and thus presents the need for more to be done. As reported, the DESI score for Czechia in terms of Human capital reduced from 47.2 in 2021 to 45.6 in 2022, likewise the integration of digital technology plunging from 39.1 in 2021 to 33.8 in 2022. In all, the total DESI ranking of Czechia in went from 18th to 19th in 2022 from 2021, confirming the need for improvement in order to catch up with their European counterparts (DESI, 2022).

Nonetheless, the findings from this segment of the research reiterates support and serves as confirmation to the implementation of fourth industrial revolution in the retail sector.

6.2.2 Summary and Discussion of the Impacts of Industry 4.0 in retail sector

This aspect of the research sought to obtain information regarding the impacts the employees perceive from the new technology implementation in their workplace. Findings from the researcher provides insight to the changes that have come with the implementation of the fourth industrial revolution – related technologies.

With regards to the changes in the work structure, 82% of the respondents were in agreement to the fact that the new technology implemented have changed and will change the way their job in the retail sector is done. However the responses were

indifferent on answering if it will make their jobs harder or easier as 63% of respondents answered neutral if the new technology will make their job harder, with only 28% affirming that new technology will make their job harder. 66% of the total respondents also confirmed that the new technology will give them less tasks to work on based on the tasks they currently have technology performing. 229 respondents confirmed this by giving examples of tasks that are performed by technology that were done by human, like pricing of goods, completing forms, invoicing, taking stock, inventory, placing orders, amongst others. A high proportion of industries in different sectors are optimistic on the growth and benefits automation and technologies will bring and are willing to employ them more and more to perform most repetitive tasks or those that do not require much of human intelligence (Yuh & West, 2001).

These impacts may result in labor substitution as up to about 14% of the global workforce would be displaced in their jobs, given that a quarter of organizations were reported to have the intention to restructure their operations with automation to increase productivity (McKinsey Global Institute, 2017). The respondents in this research expressed their opinions in the same direction as this McKinsey report, where 85% of the responses showed that the employees feel the new technologies will reduce the number of jobs in the retail sector. 69% of the respondents go on to answer that the future of working in the retail sector is at risk due to the implementation of new technologies as represented in figure 16.

Employees will now need to learn new skills to cope with the changes in the retail sector or have to learn entire new skills in order to work in a different sector should their roles be replaced with technology. In this research, 72% of the respondents reveal that they will need new skills in order to survive the changes technologies may bring to the retail industry. 151 of the respondents expressed their views on the kind of support they may need, varying from training from qualified personnel to self e-learning with the appropriate information being readily and conveniently available. Other views include constant support from management when there are new technologies and the opportunity for them to learn these skills step by step, as

well as the continuous provision of learning opportunities in order to stay sharp and in tune with every improvement or changes that may come along with the fourth industrial revolution in the retail sector. According to Goos et al. (2019), while the need for skills will be a requirement for employees to be equipped with, there might also be a situation where robotics and new technologies will make work easier, so much so, that it may actually require less skills to accomplish work.

The implementation of the fourth industrial revolution in the various sectors of the economy may influence the quality of life, as well as the work-family life balance of employees. Working with technology would allow for flexibility in the style of work, for example eliminating the fixed traditional working hours, encouraging freelancing, having project – based employment, etcetera, to enable employees plan their life outside work and prioritize it according to how they deem fit. With the absence of the same set of boundaries, employees may be expected to be available at different times of the day, making the work demand on them stochastic and hence hinder a blockade between schedule for work life, and then family life which will end affecting the work-life balance of employees with both sides of the same coin (Schröder, 2016). In this paper, 221 of the total respondents which correspond with 55.39% of them answered that they have children or dependants in their lives. This then makes it essential for them to have the needed balance to handle work responsibilities, as well as family and social responsibilities efficiently, to minimize physical strain, risk of injury and sickness, which can ultimately improve the health and well-being of the retail employees in order to improve their productivity as well. It is therefore an imperatively important factor for this research to analyze such an effect on the employees in the retail sector. 43% of the retail employees agree that the implementation of industry 4.0 will improve their work – life balance, while 51% responded indifferently which implies that they cannot ascertain whether their work-life balance will see an improvement or not. This neutrality could be associated with the assertion by about the possibility of employees being affected both positively and negatively in their work and life balance by the advancements in technology that comes with Industry 4.0 implementation.

6.3 Recommendations

Based on the findings so far, the researcher would make the following recommendations in order to counter the impacts that come along with Industry 4.0 implementation in the retail sector.

6.3.1 Acquiring Soft Skills

Even though the fourth industrial revolution is changing the nature of work into more of automation and technology inclined, not every task can or may need to be automated. Therefore, the author recommends that retail employees should seek to improve their cognitive and soft skills. Retail is more of a service sector and thus, even when the routine roles are automated, soft skills are less likely to be automated and hence employees with these skills may have the opportunity to stay in the sector. Learnability, teamwork, curiosity, adaptability, etcetera, are all forms of soft skills, such that the ability for the retail employees to learn and adapt to the new technologies implemented in their sector will become a very necessary skill to possess (Ra et al., 2019). Since soft skills can be learnt and taught, retail employees should seek to develop their skillset in this aspect to improve their chances of surviving the impacts of Industry 4.0.

6.3.2 Modification of Education Models

The impacts of the fourth industrial revolution is certainly not limited to the retail sector, but several sectors of the economy and thus the involvement of the government is needed to maintain balance. The author of this paper would like to suggest an intervention of the Czech government in implementing new educational models to help the younger generations grow into the evolution and newer technologies that may come, to make them industry ready.

In recent years, educational systems worldwide have witnessed significant integration of technology across various levels. The technological implementations span from equipping classrooms with information and communication technology, to em-

ploying scripted lesson plans and adaptive learning software tailored for personalized education. Similarly, higher education has seen a surge in online courses and blended learning approaches, including open online courses. Noteworthy success stories in educational technology include game-based learning initiatives in the People's Republic of China, linking improved language learning outcomes to advancements in mathematics achievement. Moreover, adaptive learning platforms like ASSISTments in the US and Mindspark in India have demonstrated positive impacts by catering content according to individual learning paces, supported by the underlying principle of teaching aligned with students' ability levels (Lai et al., 2015). These instances emphasize the potential for technology to enhance learning experiences and if the Czech government adopts these models into the Czech system along with the implementation of "Prumysl 4.0", it will prepare the populace for the dynamics and effects that may come with the revolution.

6.3.3 Integrated Stakeholder Engagement

The author recommends government, as well as private employment agencies to form alliance with employees and employers. When new technologies are implemented, there will be lesser tasks for employees to work on, and eventually less jobs for employees. There is also an issue mismatch between skills and technologies which need to be bridged. When employees are pushed out of jobs, these agencies can aid to match labour to available vacancies by addressing issues related to inadequate information about employees for employers, and taking appropriate actions to problems that may arise in the retail job market, in order to promote occupational mobility.

6.4 Limitation of Study

Results from this research highlighted how the fourth industrial revolution affects the retail labor but not the retail sector itself. In future, researchers can look to assess the future of the retail sector itself. Further research could look into what kind of training and skills employees would receive in order to work with new technologies, to find out if they will stay in the retail sector, or seek to apply these skills in another job instead.

6.5 Conclusions

This research sought to investigate the transformative impacts of technological advancements associated with the fourth industrial revolution on the labor market within the retail sector, focusing specifically on Brno, Czechia. With the author employing electronic questionnaires, direct insights were gathered from retail employees in Brno to explore the ramifications of these advancements. The primary research questions centred on how these technological shifts would affect retail jobs and the labor market, to what extent this evolution would shape the industry's future, and why embracing these changes in the retail labor market is imperative.

The findings underscored the profound alterations brought about by the fourth industrial revolution in the workload of retail workers. A significant majority, comprising 71.6% of respondents, acknowledged this shift. Moreover, an overwhelming 82.9% emphasized the substantial changes expected in the operational aspects of the retail market due to these technological advancements. Notably, 66.9% of respondents concurred that Industry 4.0 would diminish the tasks performed by retail employees, consequently posing a threat to their job security, a concern shared by 69% of respondents.

The research revealed a consensus among respondents (85.7%) that the fourth industrial revolution would result in fewer jobs within the retail sector, as demonstrated by the displacement of traditional roles by new technologies like invoicing,

inventory management, and others. Furthermore, the integration of new technologies will render current skill sets obsolete, necessitating retail employees to acquire new skills for relevance in their field. While 84% of respondents indicated receiving training for current technologies, 72% stressed the importance of acquiring new skills to adapt to the revolutionary changes in the retail sector.

However, the question about what extent the impact exerted by the fourth industrial revolution will impact the future of the retail industry, defies simplistic categorization due to the ongoing and continuous nature of the evolution of technological innovations. Respondents reported a surge in technology implementation, from cleaning robots and self-checkout booths to the emergence of shops devoid of human staff, reflecting a trend shifting towards automation and online retail dominance. This trajectory, however, raises concerns among employees regarding job security and the potential replacement of tasks by machines and technology.

Only 43% of respondents perceived an improvement in work-life balance resulting from technological implementations, while a significant majority remained uncertain about the overarching impacts the revolution would bring to their lives.

The implications drawn from this research highlight the undeniable and substantial influence that the fourth industrial revolution will exert on the retail labor market. Therefore, proactive measures by retail management are imperative to assist employees in embracing this revolution. The Czech government's endorsement of "Prumysl 4.0," aimed at integrating digital technologies to enhance productivity and competitiveness across all sectors of the economy, underscores the necessity for the retail sector to consider adapting and embracing these changes in order to not only enjoy the benefits but also play an instrumental role in minimizing the disruptive effects of the revolution (Adámek, 2018).

In conclusion, the fourth industrial revolution represents a formidable force poised to reshape the retail labor market significantly. Acknowledging its inevitability and aligning with strategies to mitigate its impact are critical imperatives for the retail sector to thrive in the face of this transformative era.

7 References

- Adamcova, P. (2022). Nákup nonstop a bez prodavačů. [Online] Retrieved from: <https://zpravy.aktualne.cz/finance/nakupovani/coop-samoobslužna-prodejna/r~9170ba30a69711ecb13cac1f6b220ee8/>
- Adámek, P. (2018). A research agenda of industry 4.0 from the Czech Perspective. *Digital Transformation in Smart Manufacturing*. doi: 10.5772/intechopen.71798. [Online] Retrieved from: [A Research Agenda of Industry 4.0 from the Czech Perspective | IntechOpen](#)
- Akoglu, H. (2018). User's guide to correlation coefficients. *Turkish journal of emergency medicine*, 18(3), 91-93.
- Amirrudin, M., Nasution, K., & Supahar, S. (2021). Effect of variability on Cronbach alpha reliability in research practice. *Jurnal Matematika, Statistika dan Komputasi*, 17(2), 223-230.
- Britannica, T. Editors of Encyclopaedia (2023, January 6). DNA. Encyclopedia Britannica. [Online] Retrieved from: <https://www.britannica.com/science/DNA>
- Castelo-Branco, I., Amaro-Henriques, M., Cruz-Jesus, F., & Oliveira, T. (2023). Assessing the Industry 4.0 European divide through the country/industry dichotomy. *Computers & Industrial Engineering*, 176, 108925.
- Collins, F. S., & McKusick, V. A. (2001). Implications of the Human Genome Project for medical science. *Jama*, 285(5), 540-544.
- Conrads, J., Klitou, D., Pedersen, B., Probst, L., & Rasmussen, M. (2017). Digital Transformation Monitor. *Germany: Industrie 4.0*. European Commission. [Online] Retrieved from: [Microsoft PowerPoint - DTM_Industrie 4.0 v1 \(europa.eu\)](#)
- Czech Statistical Office. (November, 2022). Statistical Yearbook of the Czech Republic – 2022. 10. LABOUR MARKET. [Online] Retrieved from: <https://www.czso.cz/documents/10180/199842475/3201982210b09.xlsx/0ade0bfa-570b-473a-b3eb-8ab5606e426a?version=1.2>

-
- Delgosha, M. S., & Hajiheydari, N. (2020). On-demand service platforms pro/anti adoption cognition: Examining the context-specific reasons. *Journal of Business Research*, 121, 180-194.
- Deloitte. (2020). The Fourth Industrial Revolution—At the Intersection of readiness and responsibility. *Deloitte Insights*.
- DESI. (2021). European Commission. Digital Economy and Society Transformation 2021, Czechia. [Online] Retrieved from: [Digital Economy and Society Index \(DESI\) 2021 | Shaping Europe's digital future \(europa.eu\)](#)
- DESI. (2022). European Commission. Digital Economy and Society Transformation 2022, Czechia. [Online] Retrieved from: [The Digital Economy and Society Index — Countries' performance in digitisation | Shaping Europe's digital future \(europa.eu\)](#)
- Dunn, G. (1992). Design and analysis of reliability studies. *Statistical Methods in Medical Research*, 1(2), 123-157.
- Export Enterprises SA. (2023). The distribution network in the Czech Republic. [Online] Retrieved from: <https://www.lloydsbanktrade.com/en/market-potential/czech-republic/distribution>
- Farrell, L., Newman, T., & Corbel, C. (2021). Literacy and the workplace revolution: a social view of literate work practices in Industry 4.0. *Discourse: Studies in the cultural politics of education*, 42(6), 898-912.
- Gibbs, R. A. (2020). The human genome project changed everything. *Nature Reviews Genetics*, 21(10), 575-576.
- Gilchrist, A. (2016). The technical and business innovators of the industrial internet. *Industry 4.0: the industrial internet of things*, 33-64.
- Goos, M., Arntz, M., Zierahn, U., Gregory, T., Gomez, S. C., Vázquez, I. G., & Jonkers, K. (2019). *The impact of technological innovation on the future of work* (No. 2019/03).
- Hudson, P. (2014). *The industrial revolution*. Bloomsbury Publishing.
- Jänicke, M., & Jacob, K. (2009). A Third Industrial Revolution? Solutions to the crisis of resource-intensive growth. *Solutions to the Crisis of Resource-Intensive Growth*. JRC Working Papers Series on Labour, Education and Technology.

-
- Klitou, D., Conrads, J., Rasmussen, M., Probst, L., & Pedersen, B., (2017). Digital Transformation Monitor. *Czech Republic: "Prumysl 4.0"*. European Commission. [Online] Retrieved from: [DTM_Prumysl 40_CZ v3.pdf \(europa.eu\)](#)
- Kusmin, K. L. (2018). Industry 4.0. *IFI8101-Information Society Approaches and ICT Processes*.
- Lansbury, R. D. (2018). The changing world of work and employment relations: a multi-level institutional perspective of the future. *Labour & Industry: a journal of the social and economic relations of work*, 28(1), 5-20.
- Lai, F., Luo, R., Zhang, L., Huang, X., & Rozelle, S. (2015). Does computer-assisted learning improve learning outcomes? Evidence from a randomized experiment in migrant schools in Beijing. *Economics of Education Review*, 47, 34-48.
- Lau, A. K. T., & Hui, D. (2002). The revolutionary creation of new advanced materials—carbon nanotube composites. *Composites Part B: Engineering*, 33(4), 263-277.
- Martín, J., Cervero, A., Mir, P., Martinez, J. A. C., Pellicer, A., & Simón, C. (2013). The impact of next-generation sequencing technology on preimplantation genetic diagnosis and screening. *Fertility and sterility*, 99(4), 1054-1061.
- McGinnis, D. (2018). What is the fourth industrial revolution? *Salesforce Blog*. <https://www.salesforce.com/blog/what-is-the-fourth-industrial-revolution-4ir/>
- McKinsey Global Institute (MGI). (2017). Independent work: Choice, necessity, and the gig economy. McKinsey & Company. <https://www.mckinsey.com/featured-insights/employment-and-growth/independent-work-choice-necessity-and-the-gig-economy>
- Momeni, F., Liu, X., & Ni, J. (2017). A review of 4D printing. *Materials & design*, 122, 42-79.
- More, C. (2002). *Understanding the industrial revolution*. Routledge.
- O'Neill, A. (2022). Statista. *Czech Republic: Distribution of gross domestic product (GDP) across economic sectors 2011 – 2021*. [Online] Retrieved from: [Czech Republic - share of economic sectors in the gross domestic product 2021 | Statista](#)

-
- Nchuchuwe, F. F., & Adejuwon, K. D. (2012). The challenges of agriculture and rural development in Africa: the case of Nigeria. *International Journal of Academic Research in Progressive Education and Development*, 1(3), 45-61.
- Poor, P., & Basl, J. (2018, October). Czech republic and processes of industry 4.0 implementation. In *Proceedings of the 29th DAAAM International Symposium* (pp. 0454-0459). Vienna: DAAAM International.
- Postelnicu, C., & Câlea, S. (2019). The fourth industrial revolution. Global risks, local challenges for employment. *Montenegrin Journal of Economics*, 15(2), 195-206.
- Property Forum. (2022). Czech retail market. [Online] Retrieved from: <https://www.property-forum.eu/news/czech-retail-market-attracts-38-new-brands-in-2021/10992>
- Ra, S., Shrestha, U., Khatiwada, S., Yoon, S. W., & Kwon, K. (2019). The rise of technology and impact on skills. *International Journal of Training Research*, 17(sup1), 26-40.
- Schröder, C. (2016). The challenges of industry 4.0 for small and medium-sized enterprises. *Friedrich-Ebert-Stiftung: Bonn, Germany*.
- Schubert, C., Van Langeveld, M. C., & Donoso, L. A. (2014). Innovations in 3D printing: a 3D overview from optics to organs. *British Journal of Ophthalmology*, 98(2), 159-161.
- Schwab, K. (2017). *The fourth industrial revolution*. Currency.
- Shiller, R. J. (2013). Finance and the good society. In *Finance and the Good Society*. Princeton University Press.
- Statista. (2018). *Annual business confidence index for the industrial sector in the European Union from 2008 to 2022*. [Online] Retrieved from: [European industrial sector business confidence index 2022 | Statista](#)
- Taylor, T. A. (2018). On-demand service platforms. *Manufacturing & Service Operations Management*, 20(4), 704-720.
- Tejada, J. J., & Punzalan, J. R. B. (2012). On the misuse of Slovin's formula. *The philippine statistician*, 61(1), 129-136.

-
- Treleaven, P., Brown, R. G., & Yang, D. (2017). Blockchain technology in finance. *Computer*, 50(9), 14-17.
- Willems, L., Agrell, P. J., & Lejeune, C. (2018). *On the Supply Chain in the Fourth Industrial Revolution*. GRIN Verlag.
- World Economic Forum (WEF). (2018). The future of jobs report 2018. World Economic Forum. [Online] Retrieved from: <https://www.weforum.org/reports/the-future-of-jobs-report-2018>
- Yıldırım, E., & Çeştepe, H. (Eds.). (2018). *Globalization, Institutions and Socio-economic Performance: Macro and Micro Perspectives*. Peter Lang.
- Yoder, D. (1926). Current definitions of revolution. *American Journal of Sociology*, 32(3), 433-441.
- Yuh, J., & West, M. (2001). Underwater robotics. *Advanced Robotics*, 15(5), 609-639.

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Attachments

A Attachments



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